

Inequalities in Mathematical Olympiads, but it's 2026

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§1 Introduction

Inequalities have played an important role in all kinds of mathematical olympiads. However, the golden era of inequalities is over. Around 2000, there were numerous IMO and IMO shortlist problems that went like this: *Let a, b, c be positive real numbers, prove that the following inequality holds.* To make things nice, the inequality was mostly a symmetric inequality, i.e. a, b and c played the same role. Also, the equality case was almost always $a = b = c = 1$.

However, those kinds of problems have become much more rare. Let it be because it was possible to have bashy solutions, like with Muirhead or because those problems just got ugly. Those *cyclic 3-var* inequalities still remain in national olympiads of some countries, mostly because they are old-fashioned or uncreative.

Recently, inequalities have become quite different: Often, they have n variables and not that symmetric anymore, but still having some nice form. For these newer inequalities, it is not about bashing or those old classic inequalities anymore, we require newer techniques.

Hence, I tried to collect some ideas and techniques. I do not mean fancy theorems, only some old ideas which have been used successfully in recent years. There aren't a lot of theorems in this script, and if there are, I focus much more on how to use it and not how to prove it. Still, I tried to cover some standard methods. Also, there are always plenty of problems you can try yourself on.

[Section 2](#) covers the basics of inequalities, starting from AM-GM. Even if you have already seen this inequality, it should be a good idea to go through this chapter, as it really tells you how to *really* use this inequality, what you might not know. The same applies for [section 3](#). You might already know Cauchy-Schwarz, but not necessarily how to use it.

However, I think, as of 2025, the most important is [section 4](#), the way of doing estimates termwise. Here, there are more irregular and more-variable inequalities, not just the very classical ones. The last three chapters are somewhat more random, while the last chapter about elementary symmetric polynomials mostly applies to classic inequalities.

Alongside lots of worked examples there are plenty of problems for you to try as well, both at the end of a chapter and at the end of the script (the letter of which does not spoil which technique is to be used). There are numbered hints, following the same principle as for example EGMO. In PDF versions, they are clickable, in printed ones of course not. However, the hints are listed in ??, starting with 1. So the hints are accessible without you getting spoiled.



Figure 1: This is Mount Inequality. Currently, it is calm, but we can never know when it erupts for the next time. However, lately, increased activity has been recorded...

I admit it is somewhat cheeky, but I decided to give lots of examples and “how to use it” instead of rigour and proofs. However, I think it is better understood with examples. Also, there is the phenomenon that in classical inequality scripts, there are a lot of fancy theorems with proofs, but no remarks what so ever how to use them in actual contest problems. Back then, that made me be able to recite the theorem, but not to use it, so I did not use them. This is what I try to avoid in this script. Also, I think that *techniques, strategies and ideas* are more important than theorems.

One thing I encountered while creating this script is that the names of the inequality solving methods are very much not standard. The same technique can have multiple names and, even worse, the same name can refer to different things. I tried to provide all names which I think apply to the given technique.

§1.1 A word about notation

I try to avoid it, but I sometimes use the notations for cyclic and symmetric sums in the following sense:

$$\sum_{cyc}^3 f(a, b, c) := f(a, b, c) + f(b, c, a) + f(c, a, b)$$

$$\sum_{sym}^6 f(a, b, c) := f(a, b, c) + f(a, c, b) + f(b, a, c) + f(b, c, a) + f(c, a, b) + f(c, b, a)$$

This notation can be extended for more than 3 variables in the obvious way. I highly recommend to track the number of summands at the top. This not only has the advantage that you are less likely to loose summands, it opens the door for some very practical abuse of notation. For four variables a, b, c, d for example, one can have

$$\sum_{sym}^6 ab = (ab + ac + ad + bc + bd + cd).$$

§2 How to use (weighted) AM-GM?

I assume that you know the famous AM-GM inequality:

Theorem 2.1 (AM-GM inequality)

Let $x_1, x_2, \dots, x_n$ be nonnegative reals. Then,

$$x_1^n + \dots + x_n^n \geq nx_1x_2 \dots x_n. \quad (1)$$

If $\alpha_1, \dots, \alpha_n$ are nonnegative reals with $N = \alpha_1 + \dots + \alpha_n$, then we can use them as weights:

$$\alpha_1 x_1^N + \dots + \alpha_n x_n^N \geq Nx_1^{\alpha_1} \dots x_n^{\alpha_n} \quad (2)$$

Equality occurs if and only if all x_i have the same value (or all but one α_i are zero).

Probably you have seen the inequality in another form, where there is actually an arithmetic and a geometric mean appearing, i. e. writing y_i instead of x_i^n . Then, AM-GM looks like this:

$$\frac{y_1 + \dots + y_n}{n} \geq \sqrt[n]{y_1 y_2 \dots y_n}$$

However, the formulation in the blue box has two advantages - first of all, AM-GM is often used in this form. Secondly, this form tells you, what AM-GM really means:

A sum of powers is larger than a product, as long as you do a fair comparison.

Example 2.2

Check how all those inequalities arise from AM-GM (all occurring variables are nonnegative):

- a) $a^2 + b^2 \geq 2ab$
- b) $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$
- c) $2a^3 + b^3 \geq 3a^2b$
- d) $a^3 + b^3 + 1 \geq 3ab$
- e) $(a+b)(a+c)(b+c) \geq 8abc$
- f) $x + \frac{1}{x} \geq 2$
- g) $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$
- h) $\frac{x^2+2}{\sqrt{x^2+1}} \geq 2$
- i) $a^2 + b^2 + c^2 \geq ab + ac + bc$ **Hints:** 3 178
- j) $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2}{\frac{1}{a}+\frac{1}{b}}$ **Hints:** 60
- k) $\sqrt{\frac{a^2+b^2+c^2}{3}} \geq \frac{a+b+c}{3} \geq \sqrt[3]{abc} \geq \frac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}$
- l) $a^2 + b^2 + c^2 + d^2 \geq \frac{(a+b+c+d)^2}{4}$

§2.1 What is a fair comparison and how to create one?

Let us get some intuition about how to use AM-GM.

§2.1.1 Equality cases

For example, assume that you have to find the minimum value of $a^2 + b$ under the constraint that $ab = 2$. You might be tempted to use $a^2 + b \geq 2\sqrt{a^2b}$. However, there is a problem with that. You want to have an inequality sign that is actually *equality*, when a and b are set as the optimal values. With a bit of trying around, the best we can get is $a = 1$, $b = 2$. But AM-GM has the equality case if all variables are equal, so you cannot just AM-GM a^2 and b . What we do instead is to split b in $\frac{b}{2}$ and $\frac{b}{2}$, giving the AM-GM estimate

$$a^2 + b = a^2 + \frac{b}{2} + \frac{b}{2} \geq 3\sqrt[3]{a^2 \cdot \frac{b}{2} \cdot \frac{b}{2}} = 3\sqrt[3]{\frac{(ab)^2}{4}} = 3.$$

This is the general pattern when solving an inequality problem. The first thing you always want to do is to find or guess the **equality case**. For *Show $A \geq B$* problems, this is just finding some set of variables such that equality occurs. Note that there might be multiple equality cases. On the other hand, if it is a minimise/maximise problem, we have to guess and have no confirmation that the equality case is the correct one. Still, it is very important to have some equality case in mind to know what type of inequality you want to use.

§2.1.2 Homogenisation

The AM-GM inequality has one more interesting property. For example, for three variables, it reads $a^3 + b^3 + c^3 \geq 3abc$. All terms have degree 3. Such expressions are called **homogenous**:

Definition 2.3. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called **homogenous of degree $\lambda \in \mathbb{R}$** if for all $t, x_1, x_2, \dots, x_n \in \mathbb{R}$,

$$f(tx_1, tx_2, \dots, tx_n) = t^\lambda f(x_1, x_2, \dots, x_n)$$

A polynomial¹ is homogenous if and only if all terms have the same degree:

Fact 2.4. Let $f(x_1, x_2, \dots, x_n)$ be a polynomial and $n \in \mathbb{N}$. Then, f is homogenous if and only if in all summands of f , the sum of exponents is n .

¹This notion can be generalised for rational functions, i.e. quotients of two polynomials. $\frac{f}{g}$ is homogenous if and only if both f and g are homogenous. If f is homogenous of degree λ and g is homogenous of degree μ , then $\frac{f}{g}$ is homogenous of degree $\lambda - \mu$.

Example 2.5

Consider the following example polynomials and not-exactly-polynomials:

- $a^3 + a^2b + abd + cde$ is homogenous of degree 3
- $a^2 + b^2 + c^2 - 4(ab + ac + bc)$ is homogenous of degree 2
- $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} + 1$ is homogenous of degree 0
- $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} - \frac{2}{a+b} - \frac{2}{a+c} - \frac{2}{b+c}$ is homogenous of degree -1
- $a + b + c - 1$ is not homogenous.
- $(a - 1)(b - 1)(c - 1)$ is not homogenous.

We call an inequality homogenous, if both sides are homogenous of the same degree. Most inequalities you will see are either homogenous, or have a non-homogenous side condition. Let us consider an example.

Example 2.6 (Exercise 2.18)

Let a, b, c be positive real numbers with $abc = 1$. Show that

$$a^2 + b^2 + c^2 \geq a + b + c$$

We cannot directly use AM-GM, as the inequality is not homogenous. So it might be a good idea to homogenise. That is, if we can prove that

$$a^2 + b^2 + c^2 \geq (a + b + c)\sqrt[3]{abc}, \quad (3)$$

we can kill the problem. We will see in a moment that we can now hope for proving (3) for general $a, b, c > 0$, not just those with $abc = 1$.

§2.1.3 Without loss of generality

Homogenous inequalities have the funny property that the inequality remains itself if all variables are replaced by the same multiple of itself. Consider for example the inequality $a^2 + b^2 + c^2 \geq (a + b + c)\sqrt[3]{abc}$. To rephrase it, the inequality holds for (a_0, b_0, c_0) if and only if it holds for (ta_0, tb_0, tc_0) . But we can spin this even further: It holds for (a_0, b_0, c_0) if and only if it holds $(1, \frac{b_0}{a_0}, \frac{c_0}{a_0})$. So, we can WLOG² assume something like $a = 1$. Or, by the same argument, we could just as well assume $a + b + c = 1$ or something else. So, in a homogenous inequality, you can add (or drop) a non-homogenous side condition and the inequality remains equivalent. For example:

$$\begin{aligned} & a^2 + b^2 + c^2 \geq (a + b + c) \text{ for } a, b, c \text{ such that } abc = 1 \\ \iff & a^2 + b^2 + c^2 \geq (a + b + c)\sqrt[3]{abc} \text{ for } a, b, c \text{ such that } abc = 1 \\ \iff & a^2 + b^2 + c^2 \geq (a + b + c)\sqrt[3]{abc} \text{ for all } a, b, c > 0 \\ \iff & a^2 + b^2 + c^2 \geq (a + b + c)\sqrt[3]{abc} \text{ for } a, b, c \text{ such that } a + b + c = 1 \\ \iff & a^2 + b^2 + c^2 \geq \sqrt[3]{abc} \text{ for } a, b, c \text{ such that } a + b + c = 1 \end{aligned}$$

²Without loss of generality - this means, the other cases work either very analogously or it can alternatively mean that the general case was reduced to the special case.

and so on.

There is a very different way to make assumptions without loss of generality. If the inequality is **symmetric**, i.e. all variables play the same role, we can assume something like $x_1 \leq x_2 \leq \dots \leq x_n$. However, if the inequality is only **cyclic**, i.e. it remains the same when you send x_1 to x_2 , x_2 to x_3 , and so on, sending x_n to x_1 , then we can only assume $x_1 \leq x_2, x_3, \dots, x_n$ (or $x_1 \geq x_2, x_3, \dots, x_n$).

Example 2.7

The following are symmetric expressions:

- $a^4 + b^4 + c^4$
- $a^2b + a^2c + b^2a + b^2c + c^2a + c^2b - 2a - 2b - 2c$

The following expressions are cyclic, but not symmetric:

- $a^3b + b^3c + c^3a$
- $(a - b)(b - c)(c - a)$
- $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$

The following expressions are neither cyclic nor symmetric:

- $a + 2b + 3c$
- $a + b^2 + c^3$

So especially, the following is invalid:

Example 2.8

Let $a, b, c > 0$ be real numbers. Prove that

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 2.$$

Wrong solution. WLOG, let $a \geq b \geq c$. Then,

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 1 + 1 + 0 = 2.$$

□

The problem with the wrong solution was of course that the inequality was only cyclic, not symmetric.

!

Warning 2.9. In general, you usually do NOT want to make WLOG assumptions. They may make the inequality *look* simpler, however, in general you want that your inequality has properties like being homogenous or symmetric. You want to homogenise, not unhomogenise.

§2.1.4 Substitutions

If the inequality is *very* unhomogenous, plugging in the side condition into the inequality might be unreasonable, as this might blow the inequality up or create some weird square roots. But, we can sometimes create new variables:

- If a, b, c are positive reals such that $abc = 1$, we can introduce the positive real numbers x, y, z such that $a = \frac{x}{y}$, $b = \frac{y}{z}$ and $c = \frac{z}{x}$. If we rephrase the inequality in terms of x, y and z , it automatically becomes homogenous of degree zero.
- The Ravi substitution: If there is a condition that a, b and c are the sides of a triangle, you can introduce the positive variables x, y and z such that $a = y + z$, $b = x + z$ and $c = x + y$.
- If $a^2 + b^2 + c^2 + 2abc = 1$, we also can have a corresponding substitution. We can use the fact that if α, β and γ are the angles in a triangle, then

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + 2 \cos \alpha \cos \beta \cos \gamma = 1$$

But there is the law of cosines. So, we can introduce x, y and z , the sides of the triangle. Then we can make the substitution

$$a = \cos \alpha = \frac{-x^2 + y^2 + z^2}{2yz}$$

$$b = \cos \beta = \frac{x^2 - y^2 + z^2}{2xz}$$

$$c = \cos \gamma = \frac{x^2 + y^2 - z^2}{2xy}$$

However, if a, b, c were assumed positive, then that condition is turned into a condition that x^2, y^2, z^2 are side-lengths of a triangle.

- You can do a substitution to simplify a side condition. For example, if there is a side condition that $\sum_{i=1}^n \frac{1}{1+a_i} = 1$, you might want to substitute $b_i = \frac{1}{1+a_i}$ to get the simpler condition $b_1 + \dots + b_n = 1$.
- You can substitute to get rid of a denominator. For example, when proving

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2},$$

you can substitute $x = b + c, y = a + c, z = a + b$ to make the denominators simpler.

§2.2 Find the correct weights

AM-GM is like going to the gym - it is only useful if you pick the correct weights.

So far, almost every AM-GM inequality we have looked at so far (i.e. [example 2.2](#)) asked us to prove that something is bigger than a product (or a constant). However, AM-GM is much more powerful.

Example 2.10

Let a, b, c be nonnegative real numbers. Prove that

$$a^3 + b^3 + c^3 \geq a^2b + b^2c + c^2a.$$

The general idea for such inequalities is to pick the summands on the right side and match them to a convex combination to left-side summands. a^2b for example consists of two a 's and one b , so we can do

$$a^3 + a^3 + b^3 \geq 3a^2b$$

and analogously

$$b^3 + b^3 + c^3 \geq 3b^2c$$

$$c^3 + c^3 + a^3 \geq 3c^2a$$

so we can sum up these inequalities to get

$$3(a^3 + b^3 + c^3) \geq 3(a^2b + b^2c + c^2a),$$

as desired.

Sometimes, you might even have to solve a system of equations to find the correct AM-GM weights.

Example 2.11 (German MO 2025/4)

Prove that for real numbers $a, b, c > 0$,

$$\frac{a^5}{b^2} + \frac{b}{c} + \frac{c^3}{a^2} > 2a.$$

The idea is that we want to have an estimate with AM-GM weights α, β and γ , where we can assume $\alpha + \beta + \gamma = 1$:

$$\alpha \frac{a^5}{\alpha b^2} + \beta \frac{b}{\beta c} + \gamma \frac{c^3}{\gamma a^2} \geq \left(\frac{a^5}{\alpha b^2} \right)^\alpha \left(\frac{b}{\beta c} \right)^\beta \left(\frac{c^3}{\gamma a^2} \right)^\gamma = \frac{1}{\alpha^\alpha \beta^\beta \gamma^\gamma} a^{5\alpha - 2\gamma} b^{\beta - 2\alpha} c^{3\gamma - \beta}$$

So, ideally, α, β and γ are solutions of the system of equations

$$\begin{aligned} 5\alpha - 2\gamma &= 1 \\ \beta - 2\alpha &= 0 \\ 3\gamma - \beta &= 0 \\ \alpha + \beta + \gamma &= 1, \end{aligned}$$

with 4 equations for only 3 variables. Fortunately, there is still a solution, namely $(\frac{3}{11}, \frac{6}{11}, \frac{2}{11})$. So, the only thing we still have to show is

$$\left(\frac{11}{3} \right)^{\frac{3}{11}} \left(\frac{11}{6} \right)^{\frac{6}{11}} \left(\frac{11}{2} \right)^{\frac{2}{11}} > 2.$$

This is just computation, feel free to carry it out.

Feel free to solve such systems of equations! Consider something similar.

Example 2.12 (Baltic Way 2024/4)

Find the largest constant $\alpha > 0$ such that

$$(x + y + z)^3 + \alpha(x^2y + y^2z + z^2x) \geq \alpha(x^2z + z^2y + y^2x) \quad (4)$$

for all nonnegative real numbers x, y, z .

The first step is to factor - the inequality we are talking about reads

$$(x + y + z)^3 \geq \alpha(x - y)(y - z)(z - x).$$

But now we see that the right side of the inequality depends only on the *differences* between x, y, z , not the variables itself. So, if we increase each variable by t , then the right side does not change and the left side increases. So, if the inequality is true, and then we replace (x, y, z) by $(x + t, y + t, z + t)$, the inequality will become *even more true*. Hence, we can WLOG assume that one of the variables is zero. For more arguments of this type, look at [section 6.4.2](#).

So, we are left with

$$(x + y)^3 \geq \alpha(x - y)(y)(-x) = \alpha xy(y - x) \quad (5)$$

and we have to maximise $\alpha > 0$ such that the above holds. You might want to use AM-GM:

$$x \cdot y \cdot (y - x) \leq \left(\frac{x + y + (y - x)}{3} \right)^3.$$

However, this does not give us $x + y$ on the right hand side, we need two more instances of x . The next best thing you can do is to use $3x$ instead of x :

$$3x \cdot y \cdot (y - x) \leq \left(\frac{3x + y + (y - x)}{3} \right)^3 = \left(2 \cdot \frac{x + y}{3} \right)^3 = \frac{8}{27}(x + y)^3,$$

proving us (5) and hence (4) for $\alpha = \frac{81}{8}$. But how do we know if this is optimal? Well, this value is optimal if there is an equality case. Okay, and what is the equality case? For this, the equality case of AM-GM has to be true, that is, $3x = y = y - x$ has to be true. But it is not hard to check that there are no such x and y . So, this AM-GM was not the optimal one.

But how do we find the optimal AM-GM? As I already told you, **don't be afraid to solve a system of equations for correct weights**. In our case, this will look like that:

$$px \cdot qy \cdot r(y - x) \leq \left(\frac{px + qy + r(y - x)}{3} \right)^3 = \left(\frac{(p - r)x + (q + r)y}{3} \right)^3 \quad (6)$$

where you want two things:

1. The right hand side of (6) should contain a multiple of $(x + y)$. This means, it should be the case that $p - r = q + r$.
2. There should be an equality case. This can occur iff $px = qy = r(y - x)$. As $px = qy$, we should have $x : y = q : p$. As everything is homogenous, we can WLOG take $x = q$ and $y = p$. So, it should be the case that $px = qy = pq = r(y - x) = r(p - q)$.

Putting it together, we have to solve the system of equations

$$\begin{aligned} p - r &= q + r \\ pq &= r(p - q) \end{aligned}$$

You can check that this has solution

$$p : q : r = 4 + 2\sqrt{3} : 2 : 1 + \sqrt{3}$$

resulting in the estimate

$$\begin{aligned} ((4 + 2\sqrt{3})x) \cdot (2y) \cdot ((1 + \sqrt{3})(y - x)) &\leq \left(\frac{(4 + 2\sqrt{3})x + 2y + (1 + \sqrt{3})(y - x)}{3} \right)^3 \\ &= \left(\frac{(3 + \sqrt{3})(x + y)}{3} \right)^3. \end{aligned}$$

Hence,

$$\left(2(4 + 2\sqrt{3})(1 + \sqrt{3}) \cdot \frac{27}{(3 + \sqrt{3})^3} \right) xy(y - x) \leq (x + y)^3,$$

so the optimal constant is

$$\begin{aligned} \alpha &= 2(4 + 2\sqrt{3})(1 + \sqrt{3}) \cdot \frac{27}{(3 + \sqrt{3})^3} \\ &= \frac{36(2 + \sqrt{3})}{\sqrt{3}(1 + \sqrt{3})^2} = 6\sqrt{3} \end{aligned} \tag{7}$$

Note that this value is optimal because there is an equality case.

Remark 2.13. When writing your solution up in contest, do *not* write up how you find the correct weights, feel free to submit a meteorite. So, you would begin with line (7). However, as “maximise” problems are always two-part problems, you should emphasise that we cannot find a better α . Also, a short remark on why it suffices to consider only $y > x$ would be probably a good idea.

§2.2.1 Variables as exponents

Sometimes, the problem authors feel funny and put variables in exponents. This will almost always mean that you will have to use the variables as AM-GM weights.

Example 2.14

Let $a, b, c, d > 0$ with $a + b + c + d = 1$. Show that

$$a^a b^b c^c d^d \leq a^2 + b^2 + c^2 + d^2$$

§2.3 Turning the table around

Occasionally, it might be useful to switch the hypothesis and the side condition. For example, if you have $a, b, c \geq 0$ such that $a^2 + b^2 + c^2 = a^3 + b^3 + c^3$ and want to prove that

$$a + b + c + abc \leq 4$$

(inequality is made up), it might be a good idea to try to use the assumption $a+b+c+abc > 4$ to prove one of the inequalities

$$\text{a) } a^2 + b^2 + c^2 < a^3 + b^3 + c^3$$

$$\text{b) } a^2 + b^2 + c^2 > a^3 + b^3 + c^3.$$

It is not hard to find out which one you will want to prove: just plug in *some* values of a, b, c such that $a + b + c + abc > 4$ and one of the inequalities will be false. Here, for $a = b = c = 998244353$, inequality b) is false. Hence, we know that we would like to prove a).

§2.4 Practise problems

Exercise 2.15. Let $a, b > 0$. Prove that

$$a^7 + b^7 \geq a^5b^2 + a^2b^5.$$

Hints: 7

Exercise 2.16. Let a, b, c be nonnegative reals. Prove that $a^3 + b^3 + c^3 \geq a^2b + b^2c + c^2a$.

Exercise 2.17. Let a, b, c be nonnegative reals. Show that

$$\left(\frac{a+b+c}{3}\right)^3 \geq \left(\frac{ab+ac+bc}{3}\right)^{\frac{3}{2}} \geq abc$$

Exercise 2.18. Let $a, b, c > 0$ with $abc = 1$. Prove that $a^2 + b^2 + c^2 \geq a + b + c$

Problem 2.19 (USAMO 2011/1). Let $a, b, c > 0$ with $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$. Show that

$$\frac{ab+1}{(a+b)^2} + \frac{ac+1}{(a+c)^2} + \frac{bc+1}{(b+c)^2} \geq 3.$$

Hints: 170

Problem 2.20. Let $a, b, c \geq 0$. Prove that

$$a^3b + b^3c + c^3a \geq abc(a + b + c)$$

Problem 2.21. Let $a, b, c > 0$. Find the minimum value of

$$\frac{a^5}{b^2c} + bc + \frac{1}{ca^3} + \frac{1}{a}$$

Hints: 112

Problem 2.22 (ELMO Shortlist 2018 A1). Let a, b, c be positive reals such that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$. Show that

$$a^abc + b^bca + c^cab \geq 27bc + 27ca + 27ab.$$

Problem 2.23 (Baltic Way 2021/2). Let a, b, c be the side lengths of a triangle. Prove that

$$\sqrt[3]{(a^2+bc)(b^2+ca)(c^2+ab)} > \frac{a^2+b^2+c^2}{2}.$$

Hints: 161 58

Problem 2.24 (Baltic Way 2019/2). For all non-negative real numbers x, y, z with $x \geq y$, prove the inequality

$$\frac{x^3 - y^3 + z^3 + 1}{6} \geq (x - y)\sqrt{xyz}.$$

Hints: 123 139

Problem 2.25 (ISL 2020/A3). Suppose that a, b, c, d are positive real numbers satisfying $(a + c)(b + d) = ac + bd$. Find the smallest possible value of

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

Hints: 115 97 67

Problem 2.26 (Türkiye TST 2022/7). Let $x, y, z > 0$. Find the minimum value of

$$xy + xz + yz + \frac{1}{x} + \frac{2}{y} + \frac{5}{z}$$

Hints: 78 144

Problem 2.27 (Abelkonkurransen 2025/4b). Determine the largest real number C such that

$$\frac{1}{x} + \frac{1}{2y} + \frac{1}{3z} \geq C$$

for all real numbers $x, y, z \neq 0$ satisfying the equation

$$\frac{x}{yz} + \frac{4y}{xz} + \frac{9z}{xy} = 24$$

Hints: 120 141 82

§2.5 The Muirhead Inequality

You might have noticed that there were quite some facts about symmetric sums. With the correct AM-GM weights, you will even be able to prove things like

$$\sum_{\text{sym}}^6 a^4 b^2 \geq \sum_{\text{sym}}^6 a^3 b^2 c$$

for example by finding $a^3 b^2 c$ as a suitable combination of $a^4 b^2$, $b^4 c^2$ and $c^4 a^2$. There is a much more general fact behind that. Let us consider symmetric sums of the form

$$[e_1, \dots, e_n] := \sum_{\text{sym}}^{n!} x_1^{e_1} \dots x_n^{e_n}$$

We say that the exponent sequence $(e_1, \dots, e_n)$ **majorises** $(f_1, \dots, f_n)$ if the following is true:

$$\begin{aligned} e_1 &\geq f_1 \\ e_1 + e_2 &\geq f_1 + f_2 \\ &\vdots \\ e_1 + \dots + e_{n-1} &\geq f_1 + f_2 + \dots + f_{n-1} \\ e_1 + e_2 + \dots + e_n &= f_1 + f_2 + \dots + f_{n-1} + f_n \end{aligned}$$

Theorem 2.28 (Muirhead Inequality)

Let $e_1, \dots, e_n, f_1, \dots, f_n$ be real numbers and $x_1, \dots, x_n$ be nonnegative real numbers and assume that $(e_1, \dots, e_n)$ majorises $(f_1, \dots, f_n)$. Then, $[e_1, \dots, e_n] \geq [f_1, \dots, f_n]$ with equality if and only if all variables are equal.



Warning 2.29. I recommend not to actually use Muirhead but to find the concrete AM-GM weights in contest, because graders might not accept it.

This inequality can be used as a bashing technique. If you have an inequality, multiply everything out and write it in terms of the symmetric sums. Then, you may try to use Muirhead to compare sums. However, I do not recommend it. Not only because it might get a painful computation but also because many inequalities are just not Muirheadable (and as of 2025, PSCs try to avoid selecting Muirheadable inequalities, at least in more advanced competitions).

An example of a not Muirheadable inequality is the Schur inequality:

Theorem 2.30 (Schur)

Let a, b, c be nonnegative reals and $r \geq 1$. Then,

$$a^r(a-b)(a-c) + b^r(b-a)(b-c) + c^r(c-a)(c-b) \geq 0.$$

Equality happens if and only if either all variables are equal or one variable is zero and the other two are equal.

In particular, for $r = 1$,

$$a^3 + b^3 + c^3 + 3abc \geq a^2b + a^2c + b^2a + b^2c + c^2a + c^2b$$

So, when bashing with Muirhead, it might be useful to have the Schur inequality in mind.

§3 How to use the Cauchy-Schwarz Inequality?

The famous Cauchy-Schwarz inequality is given in the following form:

Theorem 3.1 (Cauchy-Schwarz)

Let $x_1, \dots, x_n$ and $y_1, \dots, y_n$ be two sequences of *real numbers*. Then,

$$(x_1^2 + \dots + x_n^2)(y_1^2 + \dots + y_n^2) \geq (x_1y_1 + \dots + x_ny_n)^2.$$

Equality occurs if and only if one sequence is a multiple of the other.

Yes, this inequality has a meaning and there is a geometric intuition behind it. It has to do with the angle between two vectors in $\mathbb{R}^n$. However, for our purposes, we just don't care. A way to remember the inequality is that both sides should have degree 2 in both x and y , both should have the same number of summands and that sums of squares are intuitively larger than squares of sums. Think of AM-QM - this is exactly what you get when you set all y_i to 1.

We will not prove this inequality. The most important question isn't the proof anyway, but how to use the inequality. And the answer is: You just don't use it. We will use

Titu's lemma instead, which is essentially the same inequality, but in the form you will need it:

Theorem 3.2 (Titu's Lemma / T2 / Andreescu Theorem / Cauchy-Schwarz in Engel form)

Let $a_1, \dots, a_n$ be *real* numbers and $b_1, \dots, b_n$ be *positive real* numbers. Then,

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \dots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{b_1 + b_2 + \dots + b_n}$$

Equality occurs if and only if one sequence is a multiple of the other one.

This theorem fits into our intuition that squares are larger than mixed terms - in fact, we can think about this as a weighted variant of the QM-AM inequality.

Exercise 3.3. a) Convince yourself that Cauchy-Schwarz and Titu's lemma are the same inequality.

b) Prove Titu's lemma by induction.

The main use case of Titu's lemma is to *get rid of denominators*.

Example 3.4 (Based on Türkiye TST 1997/6)

Let $a, b, c, d, e > 0$ with $a^3 + b^3 + c^3 + d^3 + e^3 = 5$. Find the minimum value of

$$S := \frac{a^5}{b+c+d+e} + \frac{b^5}{a+c+d+e} + \frac{c^5}{a+b+d+e} + \frac{d^5}{a+b+c+e} + \frac{e^5}{a+b+c+d}$$

First of all, let us hope that the minimum occurs when all variables are equal to 1, so that we can apply Titu's lemma in an "innocent" way. The most straightforward application of Titu's lemma would be

$$S \geq \frac{(a^{2.5} + b^{2.5} + c^{2.5} + d^{2.5} + e^{2.5})^2}{4(a+b+c+d+e)}.$$

However, we have no good lower bound for $a^{2.5} + b^{2.5} + c^{2.5} + d^{2.5} + e^{2.5}$, as the condition $a^3 + b^3 + c^3 + d^3 + e^3 = 1$ only gives an upper bound. But fortunately, this can be fixed quite easily:

$$\begin{aligned} S &= \frac{a^6}{a(b+c+d+e)} + \frac{b^6}{b(a+c+d+e)} + \frac{c^6}{c(a+b+d+e)} + \frac{d^6}{d(a+b+c+e)} + \frac{e^6}{e(a+b+c+d)} \\ &\geq \frac{(a^3 + b^3 + c^3 + d^3 + e^3)^2}{a(b+c+d+e) + b(a+c+d+e) + c(a+b+d+e) + d(a+b+c+e) + e(a+b+c+d)} \\ &= \frac{1}{2(ab+ac+ad+ae+bc+bd+be+cd+ce+de)} \\ &\geq \frac{1}{20}, \end{aligned}$$

where the last inequality can be obtained by repeatedly applying the AM-GM $3ab \leq a^3 + b^3 + 1$ or any other method you wish.

Remark 3.5. When applying Titu's lemma, always feel free to expand fractions to get the powers you would like.

Let us look at one more example from the last century.

Example 3.6 (IMO 1995/2)

Let a, b, c be positive real numbers with $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(a+c)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

Again, we have an inequality with 3 variables, equality if $a = b = c = 1$ and a sum of fractions has to be larger than something else. This cries for Titu's lemma. Mixed terms are small, so we can keep them in the denominator. Let us put some even power in the numerator:

$$\begin{aligned} S &:= \frac{1}{a^3(b+c)} + \frac{1}{b^3(a+c)} + \frac{1}{c^3(a+b)} \\ &= \frac{a^{-2}}{a(b+c)} + \frac{b^{-2}}{b(a+c)} + \frac{c^{-2}}{c(a+b)} \\ &\geq \frac{\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)^2}{2(ab+ac+bc)} \\ &= \frac{\frac{(ab+ac+bc)^2}{(abc)^2}}{2(ab+ac+bc)} = \frac{ab+ac+bc}{2} \geq \frac{3}{2}(abc)^{\frac{2}{3}} = \frac{3}{2} \end{aligned}$$

There is maybe an infinite number of problems of this kind, but some of them are really old.

Exercise 3.7 (Bosnia and Herzegovina JBMO TST 2009). Show that for all positive reals a, b, c, d ,

$$\frac{1}{a} + \frac{1}{b} + \frac{4}{c} + \frac{16}{d} \geq \frac{64}{a+b+c+d}$$

Hints: 76

Problem 3.8 (Iran MO 1996 3rd round). Let a, b, c, d be positive real numbers. Show that

$$\frac{a}{b+2c+3d} + \frac{b}{c+2d+3a} + \frac{c}{d+2a+3b} + \frac{d}{a+2b+3c} \geq \frac{2}{3}.$$

Hints: 69

Problem 3.9 (<https://artofproblemsolving.com/community/c6h1304659p6967635>).

Let a, b, c be positive numbers with $abc = 1$. Show that

$$\frac{a}{2+bc} + \frac{b}{2+ac} + \frac{c}{2+ab} \geq 1$$

Problem 3.10 (Romanian TST³). Let a, b, x, y, z be positive reals. Show that

$$\frac{x}{ay+bz} + \frac{y}{az+bx} + \frac{z}{ax+by} \geq \frac{3}{a+b}.$$

³<https://artofproblemsolving.com/community/q3h3092p9235>

Problem 3.11 (USAJMO 2012/3). Let a, b, c be positive real numbers. Prove that

$$\frac{a^3 + 3b^3}{5a + b} + \frac{b^3 + 3c^3}{5b + c} + \frac{c^3 + 3a^3}{5c + a} \geq \frac{2}{3}(a^2 + b^2 + c^2).$$

Hints: 154

Problem 3.12 (Iran 2024 Round 3, Problem A1). For positive real numbers a, b, c, d such that

$$\frac{a^2}{b + c + d} + \frac{b^2}{a + c + d} + \frac{c^2}{a + b + d} = \frac{3d^2}{a + b + c}$$

prove that

$$\frac{3}{a} + \frac{3}{b} + \frac{3}{c} + \frac{3}{d} \geq \frac{16}{a + b + 2d} + \frac{16}{b + c + 2d} + \frac{16}{a + c + 2d}.$$

Hints: 128 49 12 1 158

§3.1 Denominator Nullification

This is a fundamentally different way to apply the Cauchy-Schwarz inequality. If you have different denominators, you can make them the same.

Example 3.13 (Wang/Kadaveru ([7])^a)

Let $a, b, c > 0$ such that $a + b + c = 3$. Prove the following inequality:

$$\frac{a}{a^3 + b^2 + c} + \frac{b}{b^3 + c^2 + a} + \frac{c}{c^3 + a^2 + b} \leq 1.$$

^a<https://artofproblemsolving.com/community/c6h474905p7249851>

If all denominators were the same, we could easily simplify. We can enforce this the following way: By Cauchy-Schwarz,

$$(a^3 + b^2 + c)\left(\frac{1}{a} + 1 + c\right) \geq (a + b + c)^2$$

and hence

$$\frac{a}{a^3 + b^2 + c} \leq \frac{a(\frac{1}{a} + 1 + c)}{(a + b + c)^2} = \frac{1 + a + ac}{(a + b + c)^2}$$

and the other two cyclic variations hold as well. Summing it up,

$$\frac{a}{a^3 + b^2 + c} + \frac{b}{b^3 + c^2 + a} + \frac{c}{c^3 + a^2 + b} \leq \frac{3 + (a + b + c) + (ab + ac + bc)}{(a + b + c)^2} \leq 1,$$

using the fact that $a + b + c = 3$.

Problem 3.14 (IMO 2005/3). Let x, y, z be three positive reals such that $xyz \geq 1$. Prove that

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geq 0.$$

Hints: 64 166 38 119

§3.2 The Hölder inequality

Just for the sake of this script being complete, I give you the Hölder inequality.

Theorem 3.15 (Hölder)

Let a_i, b_i, c_i be *nonnegative* reals for $1 \leq i \leq n$. Then,

$$\left(\sum_{i=1}^n a_i^3 \right) \left(\sum_{i=1}^n b_i^3 \right) \left(\sum_{i=1}^n c_i^3 \right) \geq \left(\sum_{i=1}^n a_i b_i c_i \right)^3$$

Proof. By applying Cauchy-Schwarz twice,

$$\begin{aligned} \left(\sum_{i=1}^n a_i^3 \right) \left(\sum_{i=1}^n b_i^3 \right) \left(\sum_{i=1}^n c_i^3 \right) \left(\sum_{i=1}^n a_i b_i c_i \right) &\geq \left(\sum_{i=1}^n a_i^{1.5} b_i^{1.5} \right)^2 \left(\sum_{i=1}^n a_i^{0.5} b_i^{0.5} c_i^2 \right)^2 \\ &\geq \left(\sum_{i=1}^n a_i b_i c_i \right)^4 \end{aligned}$$

□

This should strongly remind you to the inductive proof of AM-GM, if you have already seen it. Of course, we can extend Hölder to multiple sequences the same way.

§4 Termwise estimates

Not only is it arguably the most important chapter of this script, this is also a fun technique: If you have to estimate a sum, but there seems to be no nice way to do so, a good idea can be to estimate each summand. This might have multiple flavours:

§4.1 Just nudge each summand a tiny bit, until we still get something nice

Consider the following inequality from the Stone Age:

Example 4.1 (USAMO 1997/5)

Let $a, b, c > 0$. Show that

$$\frac{1}{a^3 + b^3 + abc} + \frac{1}{a^3 + c^3 + abc} + \frac{1}{b^3 + c^3 + abc} \leq \frac{1}{abc}$$

Note that equality occurs when $a = b = c$. So, we are in a “nice” case. What can we estimate on the left side? We should estimate the denominators against something *smaller*. What is large in the denominator? Our intuition is that sums and powers are large, and so is $a^3 + b^3$. We could use AM-GM, but that would be a bit too brutal. We want to estimate $a^3 + b^3$ against the next smaller term, in our case this is $ab(a + b)$. Magically, we win:

$$\begin{aligned} \frac{1}{a^3 + b^3 + abc} + \frac{1}{a^3 + c^3 + abc} + \frac{1}{b^3 + c^3 + abc} &\leq \frac{1}{ab(a + b + c)} + \frac{1}{ac(a + b + c)} + \frac{1}{bc(a + b + c)} = \\ &= \frac{1}{a + b + c} \left(\frac{c}{abc} + \frac{b}{abc} + \frac{a}{abc} \right) = \frac{1}{abc}, \end{aligned}$$

which is exactly the desired result.

Problem 4.2 (MOP 2004⁴). Let a, b, c be positive reals such that $abc = 1$. Show that

$$\frac{1}{(a+1)^2 + b^2 + 1} + \frac{1}{(b+1)^2 + c^2 + 1} + \frac{1}{(c+1)^2 + a^2 + 1} \leq \frac{1}{2}$$

Hints: [116](#) [94](#) [104](#) [19](#)

Problem 4.3 (Serbia JBMO TST 2018). Let $a, b, c > 0$. Prove that

$$\frac{\sqrt{ab}}{a+b+2c} + \frac{\sqrt{ac}}{a+2b+c} + \frac{\sqrt{bc}}{2a+b+c} \leq \frac{3}{4}$$

Hints: [100](#) [162](#) [68](#)

Problem 4.4 (ISL 2019/A2). Let $u_1, u_2, \dots, u_{2019}$ be real numbers satisfying

$$u_1 + u_2 + \dots + u_{2019} = 0 \quad \text{and} \quad u_1^2 + u_2^2 + \dots + u_{2019}^2 = 1.$$

Let $a = \min(u_1, u_2, \dots, u_{2019})$ and $b = \max(u_1, u_2, \dots, u_{2019})$. Prove that

$$ab \leq -\frac{1}{2019}.$$

Hints: [17](#) [121](#)

Problem 4.5 (Thailand MO 2025/3). Let a, b, c, x, y, z be positive real numbers such that $ay + bz + cx \leq az + bx + cy$. Prove that

$$\frac{xy}{ax + bx + cy} + \frac{yz}{by + cy + az} + \frac{zx}{cz + az + bx} \leq \frac{x + y + z}{a + b + c}$$

Hints: [9](#) [127](#)

§4.1.1 AM-GM puzzles

This is a fun type of inequality. The idea is to AM-GM a single summand against something else. These inequalities are usually in three variables, but mostly they arrive with a side condition and are hence too ugly or unclean to do with standard bashing techniques like Schurhead, SOS or uvw. The best way to demonstrate this is through an example.

Example 4.6 (Türkiye EGMO TST 2017/3)

Let $x, y, z > 0$ such that

$$\frac{xy}{z} + \frac{xz}{y} + \frac{yz}{x} \leq 3.$$

Prove that

$$\frac{x^2}{y^3} + \frac{y^2}{z^3} + \frac{z^2}{x^3} \geq \frac{x}{y} + \frac{y}{z} + \frac{z}{x}.$$

It would be great to estimate something like $\frac{xy}{z} + \frac{x^2}{y^3}$. However, no matter how we arrange two summands, we get an ugly square root. Trying a bit around, we observe the following magic AM-GM:

$$\frac{x^2}{y^3} + \frac{xz}{y} + \frac{y}{z} \geq 3\sqrt[3]{\frac{x^3 y z}{y^4 z}} = 3\frac{x}{y}.$$

⁴<https://artofproblemsolving.com/community/c6h1940869p13364503>

So, we can add up cyclically and get the following:

$$\frac{x^2}{y^3} + \frac{y^2}{z^3} + \frac{z^2}{x^3} + 3 \geq \frac{x^2}{y^3} + \frac{y^2}{z^3} + \frac{z^2}{x^3} + \frac{xy}{z} + \frac{xz}{y} + \frac{yz}{x} \geq 2 \left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x} \right) \geq \frac{x}{y} + \frac{y}{z} + \frac{z}{x} + 3,$$

where the last inequality sign follows from AM-GM.

Of course, the real difficulty when solving such a problem is to find the correct AM-GM.

Problem 4.7 (Algeria 2025/4). Prove that for all positive real numbers x, y, z such that $xy + yz + zx = 3$ we have:

$$\frac{1}{xyz} \geq \frac{2}{3}(\sqrt{x} + \sqrt{y} + \sqrt{z}) - 1.$$

Hints: 167 106

Problem 4.8 (Baltic Way 2022/4). The positive real numbers x, y, z satisfy $xy + yz + zx = 1$. Prove that:

$$2(x^2 + y^2 + z^2) + \frac{4}{3} \left(\frac{1}{x^2 + 1} + \frac{1}{y^2 + 1} + \frac{1}{z^2 + 1} \right) \geq 5.$$

Hints: 156 2

Problem 4.9 (Türkiye JBMO TST 2016/6). Prove that

$$(x^4 + y)(y^4 + z)(z^4 + x) \geq (x + y^2)(y + z^2)(z + x^2)$$

for all positive real numbers x, y, z satisfying $xyz \geq 1$. Hints: 143 59 88

Problem 4.10 (ISL 2018/A7). Find the maximal value of

$$S = \sqrt[3]{\frac{a}{b+7}} + \sqrt[3]{\frac{b}{c+7}} + \sqrt[3]{\frac{c}{d+7}} + \sqrt[3]{\frac{d}{a+7}},$$

where a, b, c, d are nonnegative real numbers which satisfy $a + b + c + d = 100$. Hints: 109 163 21 86 34 27 23

§4.2 Build your own telescope

This is the other side of the story. Assume you have to prove $A_1 + \dots + A_n \geq C$ and the official solution goes like this:

Solution. Note that $A_k \geq B_k$. Hence,

$$A_1 + \dots + A_n \geq B_1 + \dots + B_n = C$$

□

In the above, the last equality can also be something like a straightforward application of AM-GM or similar. What we did in the last section is looking at A_k and guessing what small change we could do to A_k to get B_k . Then, we hoped that the sum of all B_k magically equals C or is at least greater than or equal to C . Sometimes however, it might be a good idea to go from right to left. So, you are thinking about in which way you could write C as a sum $C = B_1 + \dots + B_n$, where B_k ideally looks *a bit similar* to A_k . In the following, I will call such a way to write $C = B_1 + \dots + B_n$ a **telescope**, even if it isn't always a telescoping sum in the traditional sense. The guess might be intuitive, or just the *simplest term that could work that comes to mind*. Maybe you will even have to find some weights. We better show this principle with an example.

Example 4.11 (Spain MO 2024/2)

Let $a_1, \dots, a_n$ be real numbers greater than one whose product is $n + 1$. Show that

$$\prod_{k=1}^n \left(\frac{1}{k^2(a_k - 1)} + 1 \right) \geq n + 1$$

What a weird problem! The condition and the inequality both seem somewhat random at first sight. To get an intuition, what we should do first is to guess the equality case - this is not obvious! In fact, it is not even that easy to come up with numbers $a_1, \dots, a_n$ all greater than one that are at least *somewhat* nice. We have to work with numbers that are *just a bit* bigger than one. The only nice-looking thing we can come up with is $a_k = \frac{k+1}{k}$. If we plug this in, we get

$$\prod_{k=1}^n \left(\frac{1}{k^2(a_k - 1)} + 1 \right) = \prod_{k=1}^n \left(\frac{1}{k^2 \left(\frac{k+1}{k} - 1 \right)} + 1 \right) = \prod_{k=1}^n \left(\frac{1}{k} + 1 \right) = \prod_{k=1}^n \frac{k+1}{k} = n + 1,$$

so this is indeed an equality case!

Okay, but we still need to solve the problem. What we have understood until now is that in the “optimal” case, the k -th factor is $\frac{k+1}{k}$. The inequality seems way too random to do something like AM-GM on all factors, hence we want to estimate each factor. Of course, it would be naive to estimate each factor against $\frac{k+1}{k}$ as they depend on a_k . So we have to **build a telescope** - what I mean with that is that we would like to find some number $p(k)$, possibly depending on a_k , such that $\frac{1}{k^2(a_k - 1)} + 1 \geq p(k)$ and $p(1)p(2)\cdots p(n) = n + 1$. For example, our first guess could be $p(k) = a_k$. Then, we would have to prove that

$$\frac{1}{k^2(a_k - 1)} + 1 \geq a_k.$$

But the problem is that, when a_k increases (it can be arbitrarily large, a priori we know nothing at all about the other factors), the left side is decreasing, while the right side is increasing. So, our hypothesis fails for a_k large. So, we want a candidate for $p(k)$ that is falling with a_k . So, we should put a_k into the denominator. Our next best guess is $p(k) = \frac{(k+1)^2}{k^2 a_k}$. Now, this should be possible for you to check:

Exercise 4.12. Let $k > 0$ and $a > 1$ be real numbers. Then, prove that

$$\frac{1}{k^2(a - 1)} + 1 \geq \frac{(k + 1)^2}{k^2 a}$$

Now, the solution is not hard:

Solution. By [exercise 4.12](#),

$$\prod_{k=1}^n \left(\frac{1}{k^2(a_k - 1)} + 1 \right) \geq \prod_{k=1}^n \frac{(k + 1)^2}{k^2 a_k} = \frac{(n + 1)^2}{n + 1} = n + 1$$

□

What did we learn? We built a telescope, so a telescoping sum, or in our case, a product. **Telescoping** is a bit generalised, we mean by that any sum or product that is easy to compute.

Let us look at another example.

Example 4.13

Let $a, b, c > 0$. Show that

$$\frac{7a^4 - 4a^3b}{2a^2 + b^2} + \frac{7b^4 - 4b^3c}{2b^2 + c^2} + \frac{7c^4 - 4c^3a}{2c^2 + a^2} \geq a^2 + b^2 + c^2.$$

This problem can be solved either forward or backward. Solving it forward means looking at the individual summands and asking the question what we could estimate them against. This is a completely feasible way to approach the problem, but here we discuss how you could guess the telescope. The weird thing is that if b gets huge, $7a^4 - 4a^3b$ might get negative. So, we are looking for something which depends on a and b , gets negative for large b and has a cyclic sum of $a^2 + b^2 + c^2$. Easiest would be $\frac{a^2+b^2}{2}$, but it doesn't get negative. So, it is credible that we could show

$$\frac{7a^4 - 4a^3b}{2a^2 + b^2} \geq 2a^2 - b^2,$$

which comes down to $3a^4 + b^4 \geq 4a^3b$, which is true.

There is one more *magic trick* - for some 3 variable inequalities, it is possible to build a telescope of the form

$$\frac{a^r}{a^r + b^r + c^r}$$

for some, possibly negative value of r . In the following, there are some problems where you can build your own telescopes.

Problem 4.14 (North Macedonia 2025/2). Let $n > 2$ be an integer, $k > 1$ a real number, and $x_1, x_2, \dots, x_n$ be positive real numbers such that $x_1 \cdot x_2 \cdots x_n = 1$. Prove that:

$$\frac{1 + x_1^k}{1 + x_2} + \frac{1 + x_2^k}{1 + x_3} + \cdots + \frac{1 + x_n^k}{1 + x_1} \geq n.$$

When does equality hold? **Hints:** [114](#) [125](#)

Problem 4.15 (Hungary-Israel Binational Competition 2003). If $x_1, x_2, \dots, x_n$ are positive real numbers, prove the inequality

$$\frac{x_1^3}{x_1^2 + x_1x_2 + x_2^2} + \frac{x_2^3}{x_2^2 + x_2x_3 + x_3^2} + \cdots + \frac{x_n^3}{x_n^2 + x_nx_1 + x_1^2} \geq \frac{x_1 + x_2 + \cdots + x_n}{3}.$$

Hints: [105](#) [16](#)

Problem 4.16 (IMO 2001/2). Let a, b, c be positive reals. Show that

$$\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ac}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq 1$$

by finding $r \in \mathbb{R}$ such that

$$\frac{a}{\sqrt{a^2 + 8bc}} \geq \frac{a^r}{a^r + b^r + c^r}.$$

Problem 4.17 (MOP 2002⁵). Let $a, b, c > 0$. Show that

$$\left(\frac{2a}{b+c}\right)^{\frac{2}{3}} + \left(\frac{2b}{a+c}\right)^{\frac{2}{3}} + \left(\frac{2c}{a+b}\right)^{\frac{2}{3}} \geq 3.$$

Hints: [87](#)

⁵<https://artofproblemsolving.com/community/c6h3514177p34107736>

Problem 4.18 (IMO 2012/2). Let $a_2, \dots, a_n$ be positive reals with product 1. Show that

$$(1 + a_2)^2(1 + a_3)^3 \cdots (1 + a_n)^n > n^n.$$

Hints: [155](#) [15](#)

Problem 4.19 (USAMO 2004/5). Let $a, b, c > 0$. Prove that

$$(a^5 - a^2 + 3)(b^5 - b^2 + 3)(c^5 - c^2 + 3) \geq (a + b + c)^3.$$

Hints: [91](#) [92](#) [44](#)

Problem 4.20 (USAJMO 2012/3). Let a, b, c be positive real numbers. Prove that

$$\frac{a^3 + 3b^3}{5a + b} + \frac{b^3 + 3c^3}{5b + c} + \frac{c^3 + 3a^3}{5c + a} \geq \frac{2}{3}(a^2 + b^2 + c^2).$$

Hints: [29](#) [102](#)

Problem 4.21 (Mexico National Olympiad 2020/6). Let $n \geq 2$ be a positive integer. Let $x_1, x_2, \dots, x_n$ be non-zero real numbers satisfying the equation

$$\left(x_1 + \frac{1}{x_2}\right) \left(x_2 + \frac{1}{x_3}\right) \cdots \left(x_n + \frac{1}{x_1}\right) = \left(x_1^2 + \frac{1}{x_2^2}\right) \left(x_2^2 + \frac{1}{x_3^2}\right) \cdots \left(x_n^2 + \frac{1}{x_1^2}\right).$$

Find all possible values of $x_1, x_2, \dots, x_n$. Hints: [47](#) [180](#)

Problem 4.22 (EGMO 2020/2). Find all lists $(x_1, x_2, \dots, x_{2020})$ of non-negative real numbers such that the following three conditions are all satisfied:

- $x_1 \leq x_2 \leq \dots \leq x_{2020}$;
- $x_{2020} \leq x_1 + 1$;
- there is a permutation $(y_1, y_2, \dots, y_{2020})$ of $(x_1, x_2, \dots, x_{2020})$ such that

$$\sum_{i=1}^{2020} ((x_i + 1)(y_i + 1))^2 = 8 \sum_{i=1}^{2020} x_i^3.$$

A permutation of a list is a list of the same length, with the same entries, but the entries are allowed to be in any order. For example, $(2, 1, 2)$ is a permutation of $(1, 2, 2)$, and they are both permutations of $(2, 2, 1)$. Note that any list is a permutation of itself.

Hints: [131](#) [171](#) [90](#) [26](#)

Problem 4.23 (Baltic Way 2024/3). Positive real numbers $a_1, a_2, \dots, a_{2024}$ are written on the blackboard. A move consists of choosing two numbers x and y on the blackboard, erasing them and writing the number $\frac{x^2 + 6xy + y^2}{x + y}$ on the blackboard. After 2023 moves, only one number c will remain on the blackboard. Prove that

$$c < 2024(a_1 + a_2 + \dots + a_{2024}).$$

Hints: [126](#)

Problem 4.24 (Türkiye JBMO TST 2015/4). Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{a}{b} + \frac{b}{c} + \frac{c}{a} + 2(a + b + c)$$

for a, b, c positive real numbers satisfying $a^2 + b^2 + c^2 + 2abc \leq 1$. Hints: [153](#) [137](#) [111](#) [50](#)

Problem 4.25 (ISL 2021/A5). Let $a_1, \dots, a_n$ be nonnegative reals with sum 1. Show that

$$\sum_{k=1}^n \frac{a_k}{1-a_k} (a_1 + \dots + a_{k-1})^2 < \frac{1}{3}.$$

Hints: 51 73 36 75

Problem 4.26 (APMO 2024/3). Let $a_1, \dots, a_n$ be positive reals. Prove that

$$\sum_{i=1}^n \frac{1}{2^i} \left(\frac{2}{1+a_i} \right)^{2^i} \geq \frac{2}{1+a_1 a_2 \dots a_n} - \frac{1}{2^n}.$$

Hints: 133 41 130 176 42

Problem 4.27 (ISL 2021/A7). Let $n \geq 1$ be an integer, and let $x_0, x_1, \dots, x_{n+1}$ be $n+2$ non-negative real numbers that satisfy $x_i x_{i+1} - x_{i-1}^2 \geq 1$ for all $i = 1, 2, \dots, n$. Show that

$$x_0 + x_1 + \dots + x_n + x_{n+1} > \left(\frac{2n}{3} \right)^{3/2}.$$

Hints: 54 134 72 165 169 11 145

§4.3 The Tangent Line Trick

This might sound fancy, but this just means **linear telescope**. If your inequality is about minimising or maximising something like $\sum_{\text{cyc}} f(a)$, then it is a natural idea to estimate $f(a)$ against the linear approximation at a_0 (where the optimum is reached), that is, something like $f(a) \geq f(a_0) + f'(a_0)(a - a_0)$. This is a telescope if minimum or maximum is reached when all variables are a_0 . Note that you only need calculus for finding the right telescope, not for your actual write-up. Here is an example:

Example 4.28 (Japan 1997, Greece 1998)

Let $a, b, c > 0$. Prove that

$$\frac{(b+c-a)^2}{(b+c)^2+a^2} + \frac{(a-b+c)^2}{(a+c)^2+b^2} + \frac{(a+b-c)^2}{(a+b)^2+c^2} \geq \frac{3}{5}$$

Proof. As the inequality is homogeneous of degree 0, we can WLOG assume that $a + b + c = 3$. We get $\frac{(b+c-a)^2}{(b+c)^2+a^2} = \frac{(3-2a)^2}{(3-a)^2+a^2}$. Let $f(x) = \frac{(3-2x)^2}{(3-x)^2+x^2}$, with $f'(x) = \frac{(4a-6)(2a-6a+9)-(2a-3)^2(2a-6+2a)}{(2a-6a+9)^2}$. Equality occurs when $a = b = c = 1$, so we are interested in $f(1) = \frac{1}{5}$ and $f'(1) = \frac{18}{25}$. So, it might be possible to prove the magic inequality

$$\frac{(3-2a)^2}{(3-a)^2+a^2} \geq \frac{1}{5} - \frac{18}{25}(a-1)$$

This is just computation. If you write this as a polynomial, you will be able to factor out $(a-1)^2$, this is because at $a = 1$, you not only have equality, but the first derivatives match as well. $\square$

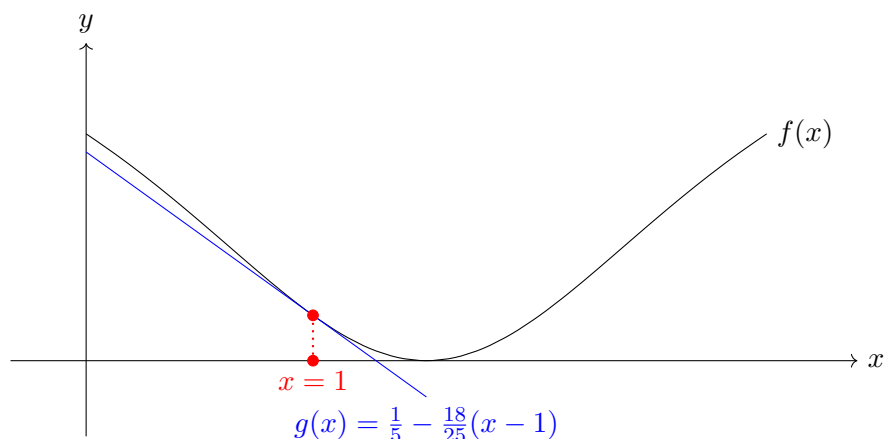


Figure 2: Even though f is not convex, $f(x)$ is just barely larger than the corresponding y -value on the tangent line. It is not a problem to use this estimate as the equality case is at $x = 1$.

Problem 4.29 (German MO 2017/5). Let x, y, z be nonnegative real numbers with $x + y + z = 1$. Prove that

$$\frac{x}{1-yz} + \frac{y}{1-xz} + \frac{z}{1-xy} \leq \frac{9}{8}.$$

Hints: 177 8

Problem 4.30 (Serbian MO 2022/2). Let a, b, c be positive reals such that $a^3 + b^3 + c^3 = 3$. Show that

$$\frac{1}{3-2a} + \frac{1}{3-2b} + \frac{1}{3-2c} \geq 3.$$

Problem 4.31 (Iran MO 2021/A1 (3rd round)). Positive real numbers a, b, c and d are given such that $a + b + c + d = 4$. Prove that

$$\frac{ab}{a^2 - \frac{4}{3}a + \frac{4}{3}} + \frac{bc}{b^2 - \frac{4}{3}b + \frac{4}{3}} + \frac{cd}{c^2 - \frac{4}{3}c + \frac{4}{3}} + \frac{da}{d^2 - \frac{4}{3}d + \frac{4}{3}} \leq 4.$$

Hints: 83

§4.3.1 The tangent line trick refined

Important: If the tangent estimate fails, it doesn't mean that you cannot solve the problem! Often, it holds almost everywhere, and where it does not hold, it consists of only edge cases very far from the optimum, which you can check manually. Here is an example of this phenomenon:

Example 4.32 (Folklore)

Let $a, b, c \geq 0$ be real numbers such that $a + b + c = 9$. Prove that

$$\frac{1}{51+a^2} + \frac{1}{51+b^2} + \frac{1}{51+c^2} \leq \frac{1}{20}$$

The inequality is already in the convenient form $f(a) + f(b) + f(c)$, where $f(x) = \frac{1}{51+x^2}$. The equality case is $x = y = z = 3$, so we are interested in $f(3)$ and $f'(3)$. We can

compute that $f(3) = \frac{1}{60}$ and $f'(3) = -\frac{1}{600}$. So, we would like to have that

$$\begin{aligned} \frac{1}{51+x^2} &\leq \frac{1}{60} - \frac{1}{600}(x-3) = \frac{13}{600} - \frac{x}{600} \\ \iff 600 &\leq (51+x^2)(13-x) = 663 - 51x + 13x^2 - x^3 \\ \iff 0 &\leq 69 - 51x + 13x^2 - x^3 = (x-3)^2(7-x) \end{aligned}$$

Oops. This is not necessarily the case. However, if all variables are at most 7, we are good. Fortunately, we have an easy fix even if one of the variables, say c , is greater than 7:

$$\frac{1}{51+a^2} + \frac{1}{51+b^2} + \frac{1}{51+c^2} \leq \frac{1}{51} + \frac{1}{51} + \frac{1}{100} < \frac{1}{50} + \frac{1}{50} + \frac{1}{100} = \frac{1}{20},$$

as desired. Always look for quick fixes if the tangent line trick “almost works”!

Try it yourself!

Problem 4.33 (MOP 2012 / Evan Chen / Romania TST 2006). Let a, b, c, d be positive integers with sum 4. Show that

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{d^2} \geq a^2 + b^2 + c^2 + d^2$$

§4.3.2 Multiple intersections

Sometimes, we only use the tangent line trick to a denominator.

Example 4.34 (USAMO 2017/6)

Let $a, b, c, d > 0$ such that $a + b + c + d = 4$. Find the minimum value of

$$\frac{a}{b^3+4} + \frac{b}{c^3+4} + \frac{c}{d^3+4} + \frac{d}{a^3+4}.$$

It might be a good idea to ponder what the equality case might be. If $a = b = c = d = 1$, then we get $\frac{4}{5}$. But in fact, we can do even better. For $a = b = 2, c = d = 0$, the sum is

$$\frac{2}{8+4} + \frac{2}{0+4} + 0 + 0 = \frac{2}{3} < \frac{4}{5}.$$

Let us try to simply put a tangent to $f(x) = \frac{1}{x^3+4}$ at $x = 2$. You can compute that it has equation $g(x) = \frac{1}{4} - \frac{1}{12}x$. Let us try to be lucky (if you want me to sound fancy, *apply wishfull thinking*) and prove $f(x) \geq g(x)$:

$$\begin{aligned} \frac{1}{x^3+4} &\geq \frac{1}{4} - \frac{1}{12}x \\ \iff 12 &\geq (x^3+4)(3-x) = 12 - 4x + 3x^3 - x^4 \\ \iff 0 &\leq 4x - 3x^3 + x^4 = x(x-2)^2(x+1) \end{aligned}$$

This is true. Note that there are actually two equality cases: When $x = 0$ and when $x = 2$. This corresponds to our observation that the equality case is $(2, 2, 0, 0)$. But now:

$$\begin{aligned} S &:= \frac{a}{b^3+4} + \frac{b}{c^3+4} + \frac{c}{d^3+4} + \frac{d}{a^3+4} \\ &\geq a \left(\frac{1}{4} - \frac{1}{12}b \right) + b \left(\frac{1}{4} - \frac{1}{12}c \right) + c \left(\frac{1}{4} - \frac{1}{12}d \right) + d \left(\frac{1}{4} - \frac{1}{12}a \right) \end{aligned}$$

$$\begin{aligned}
 &= \frac{a+b+c+d}{4} - \frac{ab+bc+cd+da}{12} \\
 &= 1 - \frac{(a+c)(b+d)}{12} \\
 &\geq 1 - \frac{((a+c)+(b+d))^2}{48} = \frac{2}{3},
 \end{aligned}$$

as desired. Equality occurs for $(a, b, c, d) = (2, 2, 0, 0)$, so we cannot do any better.

Remark 4.35. Note that our method could only have worked because the tangent $g(x)$ intersected the graph of $f(x)$ twice:

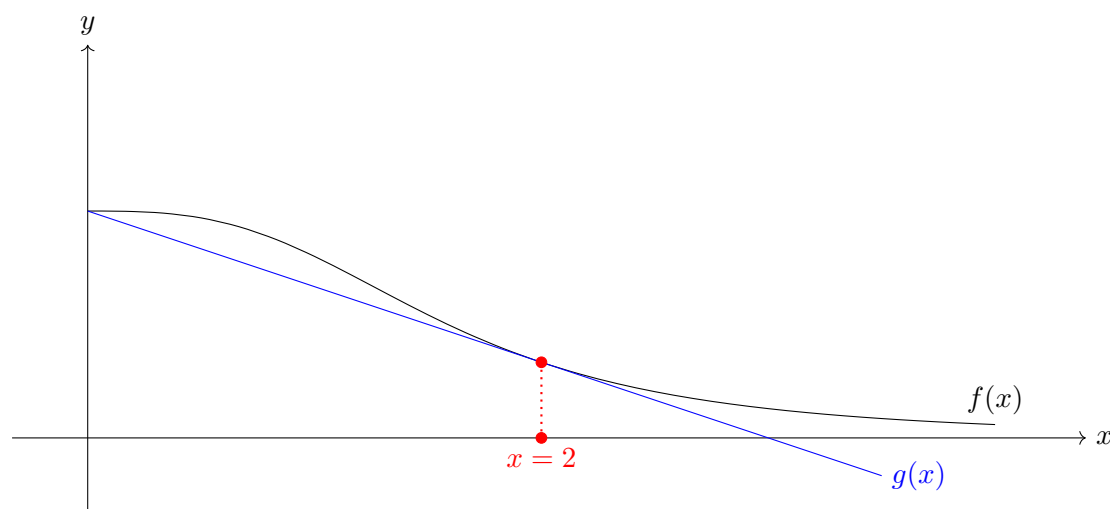


Figure 3: Conveniently, when you draw the tangent, it is below the graph of f , but there are two intersections, depicting the two equality cases.

Remark 4.36. If you can't find the equality case or you are simply not sure which tangent is the correct one, you can always try general tangents of the form $Ax + B$.

Problem 4.37 (CGMO 2007/3). Let $n \geq 4$ and $a_1, \dots, a_n \geq 0$ such that $a_1 + \dots + a_n = 2$. Prove that

$$\frac{a_1}{1+a_1^2} + \frac{a_2}{1+a_2^2} + \dots + \frac{a_n}{1+a_n^2} \geq \frac{3}{2}.$$

Hints: 31 113

§4.4 Cauchy flipping

This technique does not necessarily involve Cauchy-Schwarz or Titu's lemma. However, it is often used in combination with it.

Example 4.38 (PuMaC 2014 Individual Finals A2)

Given $a, b, c > 0$ such that $a^2 + b^2 + c^2 = 3$. Prove that

$$\frac{1}{a^3+2} + \frac{1}{b^3+2} + \frac{1}{c^3+2} \geq 1.$$

This is not a Cauchy-Schwarz problem, at least I am not aware of a Cauchy-Schwarz solution. The problem is: If you look at the denominators, they have cubes and cubes are very large. It feels like the inequality sign is pointing to the wrong direction. Whenever you feel like that, it might be a good idea to subtract each summand from something simple, e.g. a constant. In the given inequality, we can for example subtract each summand from $\frac{1}{2}$. So, we have to show that

$$\left(\frac{1}{2} - \frac{1}{a^3 + 2}\right) + \left(\frac{1}{2} - \frac{1}{b^3 + 2}\right) + \left(\frac{1}{2} - \frac{1}{c^3 + 2}\right) \leq \frac{1}{2}$$

$$\iff \frac{a^3}{a^3 + 2} + \frac{b^3}{b^3 + 2} + \frac{c^3}{c^3 + 2} \leq 1$$

which is not hard by applying the AM-GM $a^3 + 2 \geq 3a$ to each denominator.

Okay, but how did we know that $\frac{1}{2}$ was the right thing to do? Well, if you just want to stupidly apply AM-GM, it is the easiest if the numerator is also something nice. $\frac{1}{2}$ is the constant which gives you a single summand in the numerator, which you can find out by trying an arbitrary constant.

Problem 4.39 (Moldova TST 2018/6). Let a, b, c be positive reals such that $a + b + c = 3$. Prove that

$$\frac{a}{1 + b^2} + \frac{b}{1 + c^2} + \frac{c}{1 + a^2} \geq \frac{3}{2}.$$

Hints: 13 62

Problem 4.40 (Wang/Kadaveru ([7])⁶). If $a, b, c > 0$ such that $ab + ac + bc = 3$, prove that

$$\frac{a}{2a + b^2} + \frac{b}{2b + c^2} + \frac{c}{2c + a^2} \leq 1.$$

Hints: 138 22 142 37 71

§5 Integers

Sometimes, the equality cases of our inequalities cannot be attained, because there is a condition about something having to be an integer.

§5.1 Cauchy-Schwarz Approximation

Let us look at the following problem:

Example 5.1 (ISL 2024/German TSTST 2025, P5)

a) Let $x_0, x_1, x_2, \dots, x_n$ be nonnegative reals with sum n . Show that

$$2^0 x_0^2 + 2^1 x_1^2 + \dots + 2^n x_n^2 > \frac{n^2}{2}$$

b) Let $x_0, x_1, x_2, \dots, x_n$ be nonnegative integers with sum n . Find the minimum value of

$$2^0 x_0^2 + 2^1 x_1^2 + \dots + 2^n x_n^2$$

⁶<https://artofproblemsolving.com/community/c6h1413228p7950922>

For part a), which did not appear in the official shortlist, but was used at the German TSTST and is useful for an understanding, note that we can use Titu's lemma:

$$\frac{x_0^2}{2^{-0}} + \frac{x_1^2}{2^{-1}} + \cdots + \frac{x_n^2}{2^{-n}} \geq \frac{(x_0 + \cdots + x_n)^2}{1 + \frac{1}{2} + \cdots + \frac{1}{2^n}} > \frac{n^2}{2}$$

Now let us look at b). The equality case of Titu is when $x_0 : x_1 : x_2 : \cdots : x_n = 1 : \frac{1}{2} : \frac{1}{4} : \cdots : \frac{1}{2^n}$. So, we have to approximate $1 : \frac{1}{2} : \cdots : \frac{1}{2^n}$ by integers. The funny thing is that this has a nice (or niche?) real life interpretation: parliament elections. Assume that we have to fill a parliament with n delegates. Assume that the result of the election was $1 : \frac{1}{2} : \frac{1}{4} : \cdots : \frac{1}{2^n}$. That is, for example, the zeroth party got 2^n votes, the first party 2^{n-1} , the second one 2^{n-2} etc. How should we distribute the seats?

The method that is used quite often⁷ is the *Sainte-Laguë method*, also known as Webster method. This works as follows. Assume there are k parties $P_1, P_2, \dots, P_k$ and party P_i got v_i votes. Then, there exists some *divisor* q such that we can do the following: Assign

$$\left\lceil \frac{v_1}{q} \right\rceil, \left\lceil \frac{v_2}{q} \right\rceil, \dots, \left\lceil \frac{v_k}{q} \right\rceil$$

seats to the parties each, where $\lceil x \rceil$ denotes rounding to the nearest integer. This can also be interpreted as follows. We construct the following quotients:

$$\begin{array}{c} \frac{v_1}{0.5}, \frac{v_1}{1.5}, \frac{v_1}{2.5}, \frac{v_1}{3.5}, \dots \\ \frac{v_2}{0.5}, \frac{v_2}{1.5}, \frac{v_2}{2.5}, \frac{v_2}{3.5}, \dots \\ \frac{v_3}{0.5}, \frac{v_3}{1.5}, \frac{v_3}{2.5}, \frac{v_3}{3.5}, \dots \\ \vdots \\ \frac{v_k}{0.5}, \frac{v_k}{1.5}, \frac{v_k}{2.5}, \frac{v_k}{3.5}, \dots \end{array}$$

and then pick the n largest of those fractions. If we have picked r fractions for party P_i , then P_i should get r seats in the parliament. It can be shown that the two descriptions are equivalent.

A nice property of the Sainte-Laguë method is that it minimises the so-called Sainte-Laguë index: Assume the parties got $x_1, x_2, \dots, x_k$ seats and $v_1, \dots, v_k$ votes each, with $v_1 + v_2 + \cdots + v_k = v$. Then, the Sainte-Laguë index is defined as

$$S := \sum_{i=1}^k v_i \left(\frac{x_i}{v_i} - 1 \right)^2 = \sum_{i=1}^k \frac{x_i^2}{v_i} - 2 \sum_{i=1}^k x_i + \sum_{i=1}^k v_i = \sum_{i=1}^k \frac{x_i^2}{v_i} - 2n + v$$

The interpretation of this index is that it sums over each voter how disappointed they are with seat assignment. As always when considering errors, we are summing squares of differences. The fraction is compared to one as the difference between 5 and 6 seats is much more relevant than between 204 and 205 seats. And v_i is in the denominator because it is more “fixed” than x_i .

But the above calculation shows that if we set $v_i = 2^{-i}$ and $k = n + 1$, then the above problem basically asks us to show that Sainte-Laguë is optimal. Motivated by this, we

⁷e.g. Germany, Sweden, Norway, New Zealand, Bosnia and Herzegovina, Indonesia and some other countries

will be interested in the largest quotients $\frac{2^{-i}}{j-0.5} = 2\frac{2^{-i}}{2j-1} = \frac{2}{2^i(2j-1)}$ or equivalently in the smallest numbers of the form $2^i(2j-1)$. Where do we find that in the problem? Luckily, 2^i already appears. Where do odd numbers come from? Remember that $x^2 = 1 + 3 + \dots + (2x-1)$. This motivates a very short solution

Solution.

$$\begin{aligned} S &:= x_0^2 + 2x_1^2 + \dots + 2^n x_n^2 \\ &= (1 + 3 + 5 + \dots + (2x_0 - 1)) \\ &\quad + 2(1 + 3 + 5 + \dots + (2x_1 - 1)) \\ &\quad + 4(1 + 3 + 5 + \dots + (2x_2 - 1)) \\ &\quad + \dots \\ &\quad + 2^n(1 + 3 + 5 + \dots + (2x_n - 1)) \end{aligned}$$

Note that in row i we have exactly x_i summands, so in total, there are $x_0 + \dots + x_n = n$ summands. But the summands are pairwise distinct positive integers, so

$$S \geq \frac{n(n+1)}{2},$$

with equality if and only if they are the integers from 1 to n . □

Problem 5.2 (IMO 2023/4). Let $x_1, x_2, \dots, x_{2023}$ be pairwise different positive real numbers such that

$$a_n = \sqrt{(x_1 + x_2 + \dots + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right)}$$

is an integer for every $n = 1, 2, \dots, 2023$. Prove that $a_{2023} \geq 3034$.

§5.2 Secant lines

We already saw the tangent line trick for real-variabled inequalities. However, in the integer case, there is a funny tool: using a secant instead, so we might get multiple equality cases. (This is not a problem, because there are no integers between two consecutive integers.) Consider following example

Example 5.3 (Germany 2020/5 (591245))

Let $a_1, a_2, \dots, a_{22}$ be positive integers with sum 59. Prove that

$$\frac{a_1}{1+a_1} + \dots + \frac{a_{22}}{1+a_{22}} < 16$$

We could do this with the tangent line trick. However, the estimate you get from a tangent line is not strong enough. This is because of the condition of the a_i being integers. You might guess that the equality case is when the variables are close together, so some of them will be 2, some of them will be 3. This gives us the idea of an estimate against a secant. You can check that for $f(x) := \frac{x}{1+x}$, the line through $(2, f(2))$ and $(3, f(3))$ has equation $g(x) = \frac{1}{2} + \frac{x}{12}$. But note that

$$f(x) = \frac{x}{1+x} \leq \frac{1}{2} + \frac{x}{12} \iff (x-2)(x-3) \geq 0,$$

which is fulfilled for every *integer* x . Hence,

$$\frac{a_1}{1+a_1} + \cdots + \frac{a_{22}}{1+a_{22}} \leq \frac{1}{2} \cdot 22 + \frac{1}{12}(a_1 + \cdots + a_{22}) = 11 + \frac{59}{12} < 16,$$

as desired.

Problem 5.4 (Based on Korean JMO 2025/5). Let $(a_1, a_2, \dots, a_{100})$, $(b_1, b_2, \dots, b_{100})$, $(c_1, c_2, \dots, c_{100})$ all be permutations of $(1, 2, \dots, 100)$. Find the minimal value of

$$\sum_{i=1}^{100} \frac{1}{a_i + b_i + c_i}.$$

§5.3 Smoothing

This is by far the most common technique in integer-variabed inequalities - **smoothing** is just fancy terminology for pushing variables together or apart. In fact, every problem in this section can be solved by smoothing. Therefore, I only give you one more example.

Problem 5.5 (ISL 2021/A3). For each integer $n \geq 1$, compute the smallest possible value of

$$\sum_{k=1}^n \left\lfloor \frac{a_k}{k} \right\rfloor$$

over all permutations $(a_1, \dots, a_n)$ of $\{1, \dots, n\}$. **Hints:** [174](#) [140](#) [77](#) [24](#)

§6 Beyond Jensen: What you can do with functions

Here, we look at inequalities of the form

$$f(x_1) + \cdots + f(x_n) \geq \text{something}.$$

If you ever encounter an inequality of this form, or reduce your inequality to something of this form, it is quite clear what to do. This corresponds to the four levels described here. Ideally, you will apply the lowest level which already works, as in general, the later levels are more complicated.

This section will have some basic calculus in it, just a bit more than simply differentiating a function. If you don't know how to differentiate a function, there is a youtube video with a twenty-minute crash course: https://youtu.be/rTyXH8_pA.

§6.1 Level 1: Jensen and Nudging

We will use the fact that for a convex function, the chord is always above the function:

Definition 6.1. Let $I \subseteq \mathbb{R}$ be a (possibly infinite) interval and $f : I \rightarrow \mathbb{R}$ be twice differentiable. Then, f is called⁸

- **convex** if $f''(x) \geq 0$ for all $x \in I$
- **concave** if $f''(x) \leq 0$ for all $x \in I$

The Jensen inequality states that secant lines lie above a function:

⁸This is not the real definition, but sufficient for our purposes.

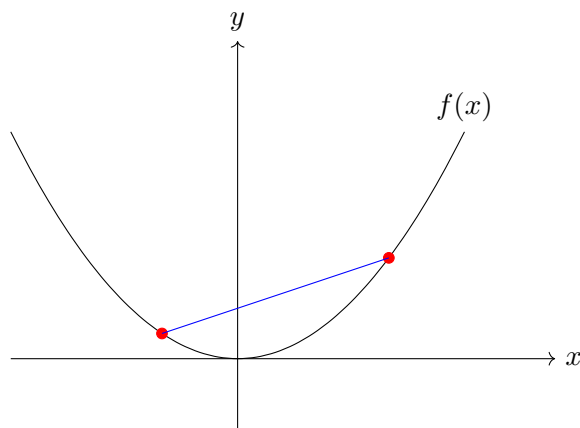


Figure 4: A convex function. It has positive second derivative, is bending to the left (above), the steepness is increasing, the area above the function is convex. The Jensen inequality for $n = 2$ states that the chord lies above the function.

Theorem 6.2 (Jensen)

Let $I \subseteq \mathbb{R}$ be as above, $f : I \rightarrow \mathbb{R}$ be convex and $x_1, x_2, x_3, \dots, x_n \in I$ and $\lambda_1, \dots, \lambda_n$ be nonnegative weights with sum 1. Then,

$$\lambda_1 f(x_1) + \dots + \lambda_n f(x_n) \geq f(\lambda_1 x_1 + \dots + \lambda_n x_n).$$

Of course, if f is concave, then $-f$ is convex, and the inequality points to the other direction.

The Jensen inequality is very general in a theoretical sense. It has AM-GM and Titu as special cases.

Exercise 6.3. Show that

- for $f(x) = \log x$, we get the AM-GM inequality.
- for $f(x) = \frac{1}{x}$, we get Titu's lemma.
- What does Jensen say for $f(x) = x^2$? Bonus question: What about $f(x) = \frac{1}{\sqrt{x}}$?

Example 6.4 (ISL 2009 A2)

Let a, b, c be positive real numbers such that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = a + b + c$. Prove that:

$$\frac{1}{(2a + b + c)^2} + \frac{1}{(a + 2b + c)^2} + \frac{1}{(a + b + 2c)^2} \leq \frac{3}{16}.$$

Our goal is to have an expression like $f(a) + f(b) + f(c)$. To achieve this, we can start by homogenising:

$$\frac{1}{(2a + b + c)^2} + \frac{1}{(a + 2b + c)^2} + \frac{1}{(a + b + 2c)^2} \leq \frac{3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)}{16(a + b + c)}.$$

Now, we can make things significantly easier if we make the WLOG assumption that

$a + b + c = 3$. Then, we just have to prove that

$$\frac{1}{(a+3)^2} + \frac{1}{(b+3)^2} + \frac{1}{(c+3)^2} \leq \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{16}.$$

So, if we define $f(x) = \frac{1}{16x} - \frac{1}{(x+3)^2}$, we have to prove that $f(a) + f(b) + f(c) \geq 0$ subject to $a + b + c = 3$. To do so, compute the second derivative of f and see that f is convex in the interval $[0, 3]$. Hence,

$$f(a) + f(b) + f(c) \geq 3f\left(\frac{a+b+c}{3}\right) = 3f(1) = 0,$$

as desired.

Example 6.5 (IMO 2001/2)

Let a, b, c be positive reals. Show that

$$\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ac}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq 1$$

Of course, this problem has quite a few solutions. One of them is to apply Jensen to the convex function $x \mapsto \frac{1}{\sqrt{x}}$. We can use a, b, c as our weights. As the inequality is homogenous of degree zero, we can WLOG assume $a + b + c = 1$.⁹ So we get

$$\begin{aligned} \frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ac}} + \frac{c}{\sqrt{c^2 + 8ab}} &\geq \frac{1}{\sqrt{a(a^2 + 8bc) + b(b^2 + 8ac) + c(c^2 + 8ab)}} = \\ &= \frac{1}{\sqrt{a^3 + b^3 + c^3 + 24abc}} \geq \frac{1}{\sqrt{1}} = 1, \end{aligned}$$

where the inequality $a^3 + b^3 + c^3 + 24abc \leq (a + b + c)^3 = 1$ can be obtained by multiplying out and then AM-GM.

Example 6.6 (<https://artofproblemsolving.com/community/c6h1304659p6967635>)

Let a, b, c be positive real numbers with $abc = 1$. Prove that

$$\frac{a}{2 + bc} + \frac{b}{2 + ac} + \frac{c}{2 + ab} \geq 1.$$

Note that $\frac{a}{2+bc} = \frac{a^2}{2a+abc} = \frac{a^2}{2a+1}$. Let $f(x) = \frac{x^2}{2x+1}$. Check that $f'(x) = \frac{2x(x+1)}{(2x+1)^2} \geq 0$ and $f''(x) = \frac{2}{(2x+1)^3} \geq 0$. Hence, f is convex and increasing, so

$$f(a) + f(b) + f(c) \geq 3f\left(\frac{a+b+c}{3}\right) \geq 3f(1) = 1$$

Actually, there are even more Jensenable problems.

Problem 6.7 (German MO 2015 3rd round (541236)). Let x, y, z be nonnegative real numbers with $x + y + z = 1$. Show that

$$\sqrt[3]{x - x^3} + \sqrt[3]{y - y^3} + \sqrt[3]{z - z^3} \leq 2$$

⁹The calculation works without this assumption as well, but the assumption makes the calculations easier.

Problem 6.8 (Evan Chen¹⁰). Let a, b, c be nonnegative reals such that $a + b + c = 3$. Prove that

$$\frac{a}{2a^2 + a + 1} + \frac{b}{2b^2 + b + 1} + \frac{c}{2c^2 + c + 1} \leq \frac{3}{4}$$

Problem 6.9 (Folklore). Let a, b, c be positive real numbers. Show that

$$\frac{a^3}{b^2(5a + 2b)} + \frac{b^3}{c^2(5b + 2c)} + \frac{c^3}{a^2(5c + 2a)} \geq \frac{3}{7}.$$

Hints: 164 179

Problem 6.10 (Israel TST 2021, Test 2, Problem 2). Let x, y, z be positive real numbers. Show that

$$\frac{x}{\sqrt{yz + 4xy + 4xz}} + \frac{y}{\sqrt{zx + 4yz + 4yx}} + \frac{z}{\sqrt{xy + 4zx + 4zy}} \geq 1.$$

Hints: 5 173

§6.2 Level 2: Tangent line trick and nudging

The tangent line trick can also be interpreted in this context. If you have written your inequality like $f(a) + f(b) + f(c) \geq L$ and Jensen simply does not work, the next best thing you can do is the Tangent Line Trick. You may want to use a refined version. However, I already wrote a section about it, so see [section 4.3](#) again.

§6.3 Level 3: The $n - 1$ equal variable principle

If the tangent line trick still does not work, there is one more useful tool. You can analyse where your function is convex and where it is concave. Then, we can use the fact that

In a convex function, pushing variables together decreases the value.
In a concave function, pulling variables apart decreases the value.

Of course, there is a more rigorous way to formulate this.

Lemma 6.11 (Smoothing)

Let $f : I \rightarrow \mathbb{R}$ convex. Let $a, b \in I$ with $a < b$ and let $0 < \varepsilon < b - a$. Then,

$$f(a) + f(b) \geq f(a + \varepsilon) + f(b - \varepsilon).$$

Similarly, if f is concave,

$$f(a) + f(b) \leq f(a + \varepsilon) + f(b - \varepsilon).$$

Proof. We prove the convex case. Apply Jensen with weights ε and $b - a - \varepsilon$:

$$f(a + \varepsilon) = f\left(\frac{a(b - a - \varepsilon) + b\varepsilon}{b - a}\right) \leq \frac{b - a - \varepsilon}{b - a}f(a) + \frac{\varepsilon}{b - a}f(b)$$

$$f(b - \varepsilon) = f\left(\frac{a\varepsilon + b(b - a - \varepsilon)}{b - a}\right) \leq \frac{\varepsilon}{b - a}f(a) + \frac{b - a - \varepsilon}{b - a}f(b)$$

Adding up those two inequalities yields the desired result. □

¹⁰<https://artofproblemsolving.com/community/c6h2010533p14091261>

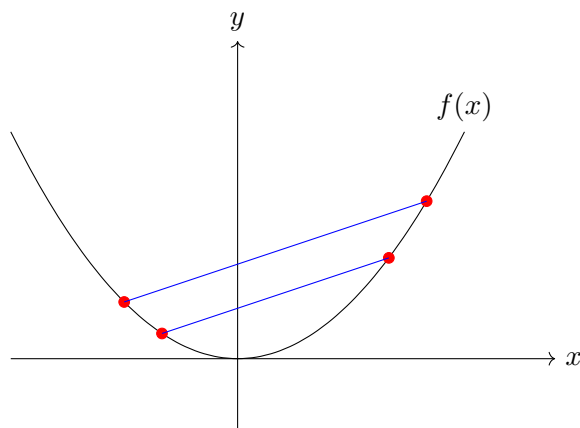


Figure 5: For a convex function, we have to push the variables together to increase the sum of the function values.

Problem 6.12 (Baltic Way 2017/1). Let $a_0, a_1, a_2, \dots$ be an infinite sequence of real numbers satisfying $\frac{a_{n-1} + a_{n+1}}{2} \geq a_n$ for all positive integers n . Show that

$$\frac{a_0 + a_{n+1}}{2} \geq \frac{a_1 + a_2 + \dots + a_n}{n}$$

holds for all positive integers n . **Hints:** 117

The above proof should remind you the way we extracted things like $a^5 + b^5 \geq a^4b + ab^4$ from AM-GM. If we go on even further, like the way we got Muirhead from AM-GM, we can get an even more general inequality, called Karamata.

The main use case of the lemma is the following: Suppose you have a function f and you want to minimise $S := f(x_1) + \dots + f(x_n)$ subject to $x_1 + \dots + x_n = s$. If f is convex, then it is just a simple application of Jensen. But what happens, if f has exactly one inflection point? Well, to minimise S , you can push together the variables in the convex part. In the concave part, however, we minimise the sum by pulling variables apart. So, if we pull two variables apart, we can do this until one of them arrives in the convex section. Then, we can push this variable together with the other ones in the convex part. Effectively, we got one variable out of the concave part. We can repeat this, until there is only one variable left in the concave part. So in total,

$n - 1$ of the variables have become equal, only making S smaller.

But then, we can make a significant generalisation of Jensen.

Theorem 6.13 ($n - 1$ equal variables principle)

Let $s \in \mathbb{R}$ be fixed. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function with exactly one inflection point. Suppose we want to minimise/maximise

$$f(x_1) + f(x_2) + \dots + f(x_n)$$

for all $x_1, x_2, \dots, x_n \in \mathbb{R}$ such that $x_1 + \dots + x_n = s$. Then, we can WLOG assume that $n - 1$ of the variables are equal.

Warning 6.14. This is a useful theorem, but be careful!

- The theorem holds only if f is defined on entire $\mathbb{R}$. This is necessary, because we want to push variables sufficiently far away in the concave part.
- You are maybe not allowed to use this theorem in this form in an actual contest. I recommend to prove the case you are using with Jensen.
- Graders prefer “move by” instead of “move until”. If you change values just once, it is clear what you mean, whereas it is not clear what “until” means on a formal level.

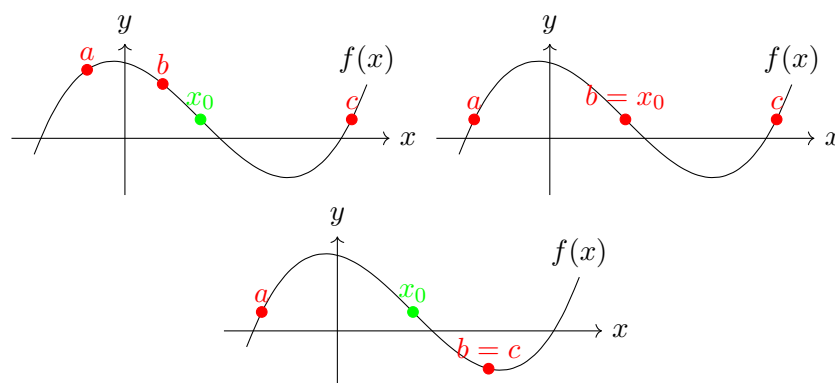


Figure 6: We want to minimise $f(a) + f(b) + f(c)$ while keeping $a + b + c$ constant. f has exactly one inflection point, namely at $x = x_0$. So, a and b are both in the concave part. Hence, if we move a and b apart, leaving $a + b$ constant, then $f(a) + f(b)$ will not increase. We can do this until we reach $b = x_0$, as you can see in the second picture. Then, we can decrease $f(b) + f(c)$ by pushing b and c together, resulting in $b = c$, as in the third picture.

Problem 6.15 (IMO 2001/2). Let a, b, c be positive reals. Show that

$$\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ac}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq 1$$

Hints: [85](#) [30](#) [132](#)

Problem 6.16 (Vietnam 1998, Mildorf). Let $x_1, x_2, \dots, x_n$ be positive reals satisfying

$$\sum_{i=1}^n \frac{1}{1998 + x_i} = \frac{1}{1998}$$

Prove that

$$\sqrt[n]{x_1 x_2 \dots x_n} \geq 1998(n - 1).$$

Hints: [32](#) [35](#)

§6.4 Smoothing, calculus and other techniques

§6.4.1 From Real Analysis: Continuous functions on compact sets

You can also freestyle a bit with functions. What happens if you increase a variable, what happens if you increase one and simultaneously decrease another (hence keeping $x_1 + x_2 + \dots + x_n$ constant) and so on. Feel free to get creative!

!

Warning 6.17. If you use calculus, make sure to **be precise!** If the derivative of a function is zero, it does not mean that it attains a minimum or a maximum there! The minimum/maximum does not have to exist! And it can be attained on the boundary, where the derivative is not necessarily zero!

However, you might find the following fact useful:

Fact 6.18. Let n be a positive (finite!) integer and let $K \subseteq \mathbb{R}^n$ be **compact**. Let $f : K \rightarrow \mathbb{R}$ be a **continuous** function.

Now you will naturally ask: What the heck do compact and continuous mean? Most importantly: You don't really have to know. In contest, you can just cite this without knowing what the words mean. Still, I try to explain it. First of all, $\mathbb{R}^n$ is just the set of n -tuples of real numbers. The other two words mean:

- continuous: Every function you will encounter is continuous, as long as it does not use $\begin{cases} a \\ b \end{cases}$. For example, you can even have functions in multiple variables. So, things like $f(a, b, c, d) = abc + \frac{ab}{d} - \frac{1}{a^2 + d^2}$ are continuous (on $K = \{(a, b, c, d) \in \mathbb{R}^4 : d > 0\}$, not on $\mathbb{R}^4$, because you cannot divide by zero!)
- compactness: A set $K \subseteq \mathbb{R}^n$ is compact, if it is **bounded** and **closed**.¹¹ Bounded means that there is $N \in \mathbb{R}$ such that K contains only tuples where each variables absolute value is at most N . Closed means that K contains its boundary. That is, roughly, that in each direction, the boundary is given by a $\geq$ or $=$ sign, not a $<$ sign.¹² Of course, $<$ signs are only a problem when you can actually get arbitrarily close to it. Otherwise, you can replace it by a $\leq$ sign.

The best way to understand is by example!

¹¹This is not the real definition. There is a more general notion of compactness, this one is only a special case. However, for our purposes (and for graders), this is enough.

¹²You can think about the “bounded” condition that we don't need any $x < \infty$ or $-\infty < x$ conditions to describe K .

Example 6.19 1. $K_1 = \{(a, b, c, d) \in \mathbb{R}^4 : 0 \leq a, b, c, d \leq 1\}$ is compact. A continuous function like $f_1(a, b, c, d) = \frac{a+b+c+d}{4} - \sqrt[4]{abcd}$ attains its maximum and minimum. (By the way, the minimum occurs e.g. for $a = b = c = d = 1$ and the maximum for $a = b = c = 1, d = 0$.) A function like $g_1(a, b, c, d) = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$ does not necessarily attain maximum and minimum, because it is not a function $K \rightarrow \mathbb{R}$. (It is not defined if one variable is zero.)

2. $K_2 = \{(a, b, c, d) \in \mathbb{R}^4 : 0 < a, b, c, d \leq 1\}$ is not compact, because it is not closed. It does not contain the part of the boundary where one of the variables is zero. You can check that g_1 is defined on K_2 , but it does not attain a maximum. However, coincidentally, it attains a minimum for $a = b = c = d = 1$.
3. $K_3 = \{(a, b, c, d) \in \mathbb{R}^4 : a, b, c, d \geq 0\}$ is not compact because it is not bounded. The variables can go off to infinity. Things like $f_3(a, b, c, d) = a + b + c + d$ do not have a maximum on K_3 .
4. $K_4 = \{(a, b, c, d) \in \mathbb{R}^4 : a, b, c, d \geq 0, a^3 + b^3 + c^3 + d^3 = 1\}$ is compact. It is bounded, because all variables have to be between 0 and 1. And it is closed, as it is only described by $\leq$ and $=$, no $<$.
5. $K_5 = \{(a, b, c, d) \in \mathbb{R}^4 : a, b, c, d \geq 0, a^2 + b^2 + c^2 + d^2 + 2abcd \leq 1\}$ is compact for the same reason.
6. $K_6 = \{(a, b, c, d) \in \mathbb{R}^4 : a, b, c, d \geq 0, a^3 + b^3 + c^3 + d^3 < 1\}$ is not compact, because it is not closed.
7. $K_7 = \{(a, b, c) \in \mathbb{R}^3 : a, b, c \geq 1, \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1\}$ is not compact, because it is not bounded. If one of the variables is just above 1, then the other ones can go off to infinity.
8. $K_8 = \{(a, b, c) \in \mathbb{R}^3 : 0 < a, b, c \leq 2, \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1\}$ is compact. First of all, note that it is bounded. Using $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$, the condition $a, b, c > 0$ can be strengthened to $a, b, c \geq 1$. Then, we don't need the $<$ sign anymore, so K_8 is compact.
9. $K_9 = \{(a, b, c) \in \mathbb{R}^3 : a^2 + b^2 + c^2 \leq 1\}$ is compact, $K'_9 = K_9 \setminus \{(0, 0, 0)\}$ is not compact, because it is not closed. The function $f_9(a, b, c) = \frac{1}{a^2 + b^2 + c^2}$ cannot be defined on K_9 , only on K'_9 . There, it attains no maximum, which is not a surprise, because K'_9 is not compact. The function

$$g_9(a, b, c) = \begin{cases} 0 & a = b = c = 0 \\ \frac{1}{a^2 + b^2 + c^2} & \text{otherwise} \end{cases}$$

can be defined on K_9 . However, it does not attain a maximum - it is not continuous!

And how does it look in the practice, how can we apply this principle? When you have a minimisation/maximisation problem involving a continuous function defined on a compact domain, you always know that the minimum/maximum is actually attained. If it happens in the interior, then all derivatives have to be zero. Otherwise, you only have to consider the boundary. Let us look at an example from half a century ago:

Example 6.20 (USAMO 1980/5)

Show that for nonnegative reals $a, b, c \leq 1$,

$$\frac{a}{b+c+1} + \frac{b}{a+c+1} + \frac{c}{a+b+1} + (1-a)(1-b)(1-c) \leq 1.$$

How do we attack such a problem? We see that the domain is not only compact, but also a cube. Compact cubes are the nicest domains. So let $K := \{(a, b, c) \in \mathbb{R}^3 : 0 \leq a, b, c \leq 1\}$ and $f : K \rightarrow \mathbb{R}$ be our function:

$$f(a, b, c) := \frac{a}{b+c+1} + \frac{b}{a+c+1} + \frac{c}{a+b+1} + (1-a)(1-b)(1-c)$$

We differentiate f with respect to each variable separately.¹³:

$$\frac{\partial f}{\partial a} = \frac{1}{b+c+1} - \frac{b}{(a+c+1)^2} - \frac{c}{(a+b+1)^2} - (1-b)(1-c)$$

But wait, this is an *increasing* function of a ! Hence, f is convex in a . Convex functions attain their *maximum* on the boundary. (More precisely: For fixed b, c , $g(a) = f(a, b, c)$ is a convex function. Jensen's inequality shows that $g(a) \leq \max\{g(0), g(1)\}$.) So we can WLOG assume $a \in \{0, 1\}$. But we can do the same for b and c ! So, at the end, we are only left with 8 cases, in which each of the variables is either 0 or 1.

Warning 6.21. In your write-up in contest, you should be careful with formulations such as “WLOG $a = 0$ or $a = 1$ ”. It is better to choose a write-up like this: *Let a, b, c be such numbers. Then, there exist $\varepsilon_1, \varepsilon_2, \varepsilon_3 \in \{0, 1\}$ such that*

$$f(a, b, c) \leq f(\varepsilon_1, b, c) \leq f(\varepsilon_1, \varepsilon_2, c) \leq f(\varepsilon_1, \varepsilon_2, \varepsilon_3),$$

because [...].

Remark 6.22. Effectively, we made use of the fact that the faces (and the edges) of the cube are again compact. By only shouting “continuous on compact”, we could only have made the WLOG assumption that one of the variables is 0 or 1. Luckily, we could do the trick with the compact domain on the boundary again, so we could reduce the number of variables from 3 to 0 and not only to 2.

§6.4.2 Differences

This fact is very simple, but I decided to dedicate an entire section to it:

If (a, b, c) is replaced by $(a+t, b+t, c+t)$, then the pairwise differences $a-b, a-c, b-c$ do not change.

This means, you can replace (a, b, c) by $(a+t, b+t, c+t)$ and set $t \in \mathbb{R}$ so that something interesting happens.

¹³Again, that funny differential sign is not Icelandic; it means that we differentiate with respect to a and treat b and c as constants.

Problem 6.23 (Austria 2025/1). Let a, b and c be pairwise distinct nonnegative real numbers. Prove that

$$(a + b + c) \left(\frac{a}{(b - c)^2} + \frac{b}{(c - a)^2} + \frac{c}{(a - b)^2} \right) > 4.$$

Problem 6.24 (IMO 2021/2). Show that the inequality

$$\sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i - x_j|} \leq \sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i + x_j|}$$

holds for all real numbers $x_1, \dots, x_n$. **Hints:** 101 45 6 160

§6.4.3 Problems

Here are some problems to practice miscellaneous calculusy methods.

Problem 6.25 (ISL 2016/A1).¹⁴ Let a, b, c be positive real numbers such that $\min(ab, bc, ca) \geq 1$. Prove that

$$\sqrt[3]{(a^2 + 1)(b^2 + 1)(c^2 + 1)} \leq \left(\frac{a + b + c}{3} \right)^2 + 1.$$

Hints: 14 18 84 181 96

Problem 6.26 (Folklore). Let $x_1, x_2, \dots, x_n \in [0, 1]$ such that $x_1 + \dots + x_n = 1$. Find the maximum value of

$$\sum_{k=1}^n x_k^4 - x_k^5.$$

Hints: 147 93

Problem 6.27. Let $n \geq 4$ and $x_1, x_2, \dots, x_n$ be nonnegative real numbers. Prove that

$$a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n + a_n a_1 \leq \frac{1}{4} (a_1 + \dots + a_n)^2$$

Hints: 61 25 33

Problem 6.28 (German MO state round 2022 (611232)). Let x, y, z be three pairwise different positive real numbers. Prove that

$$\left(\frac{x}{y - z} \right)^2 + \left(\frac{y}{x - z} \right)^2 + \left(\frac{z}{x - y} \right)^2 > 2$$

Problem 6.29 (German MO 2025/4). Prove that for real numbers $a, b, c > 0$,

$$\frac{a^5}{b^2} + \frac{b}{c} + \frac{c^3}{a^2} > 2a.$$

Hints: 108

Problem 6.30 (German MO 2016/6). Let $x_1, \dots, x_7$ be nonnegative real numbers with sum 1. Find the maximal value of

$$x_1 x_2 x_4 + x_2 x_3 x_5 + x_3 x_4 x_6 + x_4 x_5 x_7 + x_5 x_6 x_1 + x_6 x_7 x_2 + x_7 x_1 x_3.$$

Hints: 151 4 124 168 39 150 129

¹⁴This problem is way too hard to be A1.

§7 Elementary symmetric polynomials

To be honest, I only added this section for completeness and because of the fact that back then, I liked to bash old-fashioned 3-variable inequalities like this. The main use of elementary symmetric polynomials is indeed to kill symmetric 3-variable inequalities. In my opinion, in practice, they work much better than multiplying out (“Muirhead bashing”, “Chinese Dumbass Notation”, etc.), because there are plenty of inequalities for elementary symmetric polynomials, while lots of inequalities are not muirheadable or multiplying out can take too long. Also, the uvw method is a nice extension and is based on the use of elementary symmetric polynomials as well.

What are elementary symmetric polynomials, or ESP for short? They are some polynomials in n variables. For $n = 2$, there are two of them:

$$\begin{aligned}\sigma_1 &= a + b \\ \sigma_2 &= ab\end{aligned}$$

For $n = 3$, which is by far the most common case, the ESP look like this:

$$\begin{aligned}\sigma_1 &= a + b + c \\ \sigma_2 &= ab + ac + bc \\ \sigma_3 &= abc\end{aligned}$$

We can go up one more variable to see the pattern clearly. For $n = 4$, it looks like this:

$$\begin{aligned}\sigma_1 &= a + b + c + d \\ \sigma_2 &= ab + ac + ad + bc + bd + cd \\ \sigma_3 &= abc + abd + acd + bcd \\ \sigma_4 &= abcd\end{aligned}$$

So, it should have become quite obvious how this works. In n variables, σ_k is the sum of all $\binom{n}{k}$ ways to multiply k of the n variables together. The reason why they are interesting and powerful is the fact that we can express all symmetric polynomials in $\sigma_1, \dots, \sigma_n$:

Theorem 7.1 (Fundamental theorem of symmetric polynomials)

Let $P(x_1, \dots, x_n)$ be a **symmetric** polynomial of n variables, i.e. changing the values of two variables doesn't change the value of $P(x_1, \dots, x_n)$. Then, $P(x_1, \dots, x_n) = Q(\sigma_1, \dots, \sigma_n)$ for some polynomial $Q(y_1, \dots, y_n)$ in n variables.

Example 7.2

The following are symmetric polynomials in 3 variables:

- $\sigma_1, \sigma_2, \sigma_3$
- $a^2b^2c^2 + a^2b^2 + b^2c^2 + c^2a^2 - 2a - 2b - 2c - 3$
- $a^2b + a^2c + b^2a + b^2c + c^2a + c^2b$

And the following are not symmetric:

- $a + b - c$
- $a^2b + b^2c + c^2a$

We will not prove [theorem 7.1](#) here - the important thing is to find the concrete Q for your concrete polynomial, which will be something like $a^3 + b^3 + c^3$. Let us do some examples to get a feeling for it. The easiest example is $a^2 + b^2 + c^2$. To get something like a^2 without a b or c (i.e. a non-mixed term), the only way to do so is to use σ_1 , as σ_2 and σ_3 both consist of mixed terms. To get the correct degree, we have to square σ_1 . Doing so, we get:

$$\sigma_1^2 = (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2ca = (a^2 + b^2 + c^2) + 2\sigma_2,$$

hence $a^2 + b^2 + c^2 = \sigma_1^2 - 2\sigma_2$. Analogously, for $a^3 + b^3 + c^3$, we cube σ_1 , getting

$$\sigma_1^3 = (a^3 + b^3 + c^3) + 3(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b) + 6abc,$$

where $\sum_{\text{sym}} a^2b$ looks like $a \cdot ab$, so we try $\sigma_1\sigma_2$:

$$\sigma_1\sigma_2 = (a + b + c)(ab + ac + bc) = (a^2b + a^2c + b^2a + b^2c + c^2a + c^2b) + 3abc,$$

hence

$$\sigma_1^3 = (a^3 + b^3 + c^3) + 3(\sigma_1\sigma_2 - 3\sigma_3) + 6\sigma_3 = (a^3 + b^3 + c^3) + 3\sigma_1\sigma_2 - 3\sigma_3$$

giving

$$a^3 + b^3 + c^3 = \sigma_1^3 - 3\sigma_1\sigma_2 + 3\sigma_3.$$

Try yourself!

Exercise 7.3. Write those in terms of ESP:

- $a^2b^2 + a^2c^2 + b^2c^2$
- $(a + b - c)(a - b + c)(-a + b + c)$
- $(a + b)(a + c)(b + c)$

By the way, elementary symmetric polynomials also appear in a very different context. If you multiply out a polynomial that is already factored in linear factors, it looks like this:

$$(t - a_1)(t - a_2)(\cdots)(t - a_n) = t^n - \sigma_1 t^{n-1} + \sigma_2 t^{n-2} \mp \cdots + (-1)^n \sigma_n$$

This means, if the ESP are the coefficients of a polynomial, then the zeroes are exactly the variables of the inequality!

Fact 7.4. Let $a_1, \dots, a_n, c_1, \dots, c_n$ be real numbers. Consider the polynomial

$$P(t) = t^n - c_1 t^{n-1} + c_2 t^{n-2} \mp \dots + (-1)^n c_n.$$

Then, $a_1, \dots, a_n$ are the zeroes of the polynomial (counted with multiplicity) if and only if $\sigma_1 = c_1, \dots, \sigma_n = c_n$.

There are some occasions where this is useful, mostly when you already know a lot of the ESP, so that you can write down the polynomial with zeros $x_1, \dots, x_n$ quite well.

Problem 7.5. Let x, y, z be nonzero real numbers with $x + y + z = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$. Show that

$$\frac{x + y - xy}{\sqrt{1 + (y + z)^2(x + z)^2}} + \frac{x + z - xz}{\sqrt{1 + (y + z)^2(x + y)^2}} + \frac{y + z - yz}{\sqrt{1 + (x + y)^2(x + z)^2}} > 2$$

Hints: [152](#) [122](#) [103](#)

Problem 7.6 (German MO 2023 state round (621231)). The real numbers a, b, c, d satisfy

$$\begin{aligned} a + b + c + d &= 0 \\ a^3 + b^3 + c^3 + d^3 &= 0 \end{aligned}$$

Show that two of the variables add up to zero. Hints: [80](#) [46](#)

Problem 7.7 (Spain MO 2024 P4). Let a, b, c, d be real numbers satisfying

$$abcd = 1 \quad \text{and} \quad a + \frac{1}{a} + b + \frac{1}{b} + c + \frac{1}{c} + d + \frac{1}{d} = 0.$$

Prove that at least one of the numbers ab, ac, ad equals -1 . Hints: [10](#) [95](#)

Problem 7.8 (German MO 2019/6). Find all triples of positive real numbers (x, y, z) such that

$$\begin{aligned} x + \frac{y}{z} &= 2 \\ y + \frac{z}{x} &= 2 \\ z + \frac{x}{y} &= 2 \end{aligned}$$

Hints: [175](#) [172](#) [28](#)

Problem 7.9 (USAMO 2018/1). Let a, b, c be positive real numbers such that $a + b + c = 4\sqrt[3]{abc}$. Prove that

$$2(ab + bc + ca) + 4\min(a^2, b^2, c^2) \geq a^2 + b^2 + c^2.$$

Hints: [40](#) [52](#) [81](#)

§7.1 Inequalities in elementary symmetric polynomials

Suppose we are trying to prove some symmetric inequality in a, b, c . For example, the AM-HM inequality:

$$\frac{a + b + c}{3} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} = \frac{3abc}{ab + ac + bc},$$

which we already wrote in terms of ESP. So we have to prove $\sigma_1\sigma_2 \geq 9\sigma_3$. There is a slightly more convenient notation: Let us consider symmetric averages. That is:

$$d_1 := \frac{a+b+c}{3}, \quad d_2 := \frac{ab+ac+bc}{3}, \quad d_3 := abc.$$

This has the advantage that for $a = b = c = 1$, $d_1 = d_2 = d_3 = 1$. So, if there is an equality case if all variables are equal, then both sides should have the same number of terms. For example, for AM-HM, we have to prove $d_1d_2 \geq d_3$, where both sides have 1 term. (As opposed to something like $d_1d_2 \geq 2d_3$, which fails for $a = b = c = 1$, because the right side has 2 terms.) Of course, it is not hard to generalise this notation for any $n > 1$: set $d_k := \frac{\sigma_k}{\binom{n}{k}}$.

But what do we know about the d_k ? In the following, we will assume that a, b, c are nonnegative real numbers. Then, there are some inequalities:

Theorem 7.10 (Maclaurin's Inequality)

$$d_1 \geq \sqrt{d_2} \geq \cdots \geq \sqrt[n]{d_n}$$

Proof for $n = 3$. Multiply out. □

Theorem 7.11 (Newton's Inequality)

For $1 < k < n$,

$$d_k^2 \geq d_{k-1}d_{k+1}$$

Proof for $n = 3$. Multiply out. □

Theorem 7.12 (Schur's inequality)

For $n = 3$, $3d_1^3 + d_3 \geq 4d_1d_2$

Proof. Multiply out and see that it is equivalent to Schur in its classical form:

$$a^3 + b^3 + c^3 + 3abc \geq a^2b + a^2c + b^2a + b^2c + c^2a + c^2b$$

□

The Maclaurin inequality makes a lot of sense: It follows the fact that sums and powers are bigger than products. d_1 is “very sum”, d_2 is “a bit sum, a bit product” and d_3 is “very product”. The roots ensure that the comparison is fair. Equality occurs if and only if all variables are equal.

The Newton inequality doesn't have a that clear meaning. For $n = 3$, the only statement is $d_2^2 \geq d_1d_3$. This is in fact quite useful, as Maclaurin doesn't tell us which side should be larger, so it is a good addition to Maclaurin.

The Schur inequality is the only statement that says that d_3 is bigger than something. d_1^3, d_1d_2 and d_3 the only possible polynomials of degree 3, it makes sense that for sufficiently large R , we will have a statement like $Rd_1^3 + d_3 \geq (R+1)d_1d_2$. $R = 3$ heuristically makes sense because d_1^3 already has a denominator of 3^3 .

Together, these inequalities are already quite powerful:

Exercise 7.13. Sort the following by size:

- a) d_3, d_1^3, d_1d_2
- b) $d_1d_3, d_1^4, d_2^2, d_1^2d_2$
- c) $d_2d_3, d_1^5, d_1d_2^2, d_1^2d_3, d_1^3d_2$

Problem 7.14 (USAMO 1979/3). Let a, b, c be nonnegative real numbers with sum 1. Prove that

$$a^3 + b^3 + c^3 + 6abc \geq \frac{1}{4}$$

Hints: 66 110

Problem 7.15. Let a, b, c, d be positive real numbers such that $2(a + b + c + d) \geq abcd$. Prove that

$$a^2 + b^2 + c^2 + d^2 \geq abcd.$$

Hints: 65 159

§7.2 Bashing with elementary symmetric polynomials

ESP give rise to a quite powerful bashing method: If you have a symmetric, 3-variable inequality with symmetric equality case, write it in terms of elementary symmetric polynomials and use the above inequalities to finish. When can this *possibly* work?

- First of all, the inequality has to be *symmetric*. Not just cyclic, symmetric. This is because the ESP are themselves symmetric. You can only write symmetric things in terms of other things that are symmetric.
- You should be able to write the inequality as a polynomial. Of course, when exponentials and logarithms appear, you have no chance, but this is usually not the case. Fractions are usually not a problem, they can be eliminated (they can still make the calculations very complicated). If, however, the inequality is full of square roots, like for example

$$\sqrt{a^2 + b} + \sqrt{a^2 + c} + \sqrt{b^2 + a} + \sqrt{b^2 + c} + \sqrt{c^2 + a} + \sqrt{c^2 + b} \geq 2(a + b + c) + \sqrt{a + b + c},$$

(inequality is made-up) we will not be able to eliminate all the square roots. (In general, up to 3 square roots can be eliminated by repeated squaring.)

- Ideally, the equality case is when all variables are equal. There are some exceptions, where you can still apply this bashing method, even though there is no symmetric equality case:
 - The inequality is simply not sharp, there are no equality cases.
 - There is an equality case where two variables are equal, the third one is zero. Then, the only things we can do is to either use Schur's inequality, drop terms containing d_3 or to use estimates like $d_2d_3 \leq d_1^2d_3$, where both sides have d_3 and hence disappear when one of the variables is zero.

However, we mostly wish that there is a symmetric equality case. When the equality case is weird, we can still invent new inequalities using the uvw method, which will be covered in [Section 7.3](#).

- The inequality will not get way too complicated. This speaks for itself.

It is best if we look at an example.

Example 7.16 (German MO 2017/5)

Let x, y, z be nonnegative real numbers with $x + y + z = 1$. Show that

$$\frac{x}{1-yz} + \frac{y}{1-xz} + \frac{z}{1-xy} \leq \frac{9}{8}.$$

Fortunately, this is a symmetric inequality with a symmetric equality case at $x = y = z = \frac{1}{3}$, so we are on a good way to go. So, let us simply multiply out:

$$\begin{aligned} & \frac{x}{1-yz} + \frac{y}{1-xz} + \frac{z}{1-xy} \leq \frac{9}{8} \\ \iff & 8 \sum_{\text{cyc}}^3 x(1-xy)(1-xz) \leq 9(1-xy)(1-xz)(1-yz) \\ \iff & 8\sigma_1 - 8 \sum_{\text{sym}}^6 x^2y + 8\sigma_2\sigma_3 \leq 9 - 9\sigma_2 + 9\sigma_1\sigma_3 - \sigma_3^2 \\ \iff & 8\sigma_1 - 8(\sigma_1\sigma_2 - 3\sigma_3) + 8\sigma_2\sigma_3 \leq 9 - 9\sigma_2 + 9\sigma_1\sigma_3 - 9\sigma_3^2 \end{aligned}$$

But we know that $\sigma_1 = 1$! All inequalities we know about ESP are homogenous, so let us homogenise. What the problem *really* wants us to show is that

$$\begin{aligned} & 8\sigma_1^6 - 8\sigma_1^4\sigma_2 + 24\sigma_1^3\sigma_3 + 8\sigma_1\sigma_2\sigma_3 \leq 9\sigma_1^6 - 9\sigma_1^4\sigma_2 + 9\sigma_1^3\sigma_3 - 9\sigma_3^2 \\ \iff & 0 \leq \sigma_1^6 - \sigma_1^4\sigma_2 - 15\sigma_1^3\sigma_3 - 8\sigma_1\sigma_2\sigma_3 - 9\sigma_3^2 \\ \iff & 0 \leq 729d_1^6 - 243d_1^4d_2 - 405d_1^3d_3 - 72d_1d_2d_3 - 9d_3^2 \end{aligned}$$

At this point, it is not uncommon that three or even four-digit numbers appear. You should not be afraid of them! At this point, you have a perfect opportunity to check if your calculation was correct: If so, the equality case $a = b = c = d_1 = d_2 = d_3 = 1$ has remained. (It is an equality case of the homogenised inequality.) So you can check that

$$729 = 243 + 405 + 72 + 9,$$

which is true, so presumably we made no mistake during calculation. And now, it is not hard to prove the inequality:

$$\begin{aligned} 243d_1^6 & \geq 243d_1^4d_2 \\ 405d_1^6 & \geq 405d_1^3d_3 \\ 72d_1^6 & \geq 72d_1d_2d_3 \\ 9d_1^6 & \geq 9d_3^2, \end{aligned}$$

all by Maclaurin. Summing up those inequalities gives the desired result.

§7.3 The *uvw* method (Mixing Variables)

Let us try one more example.

Example 7.17 (Iran 1996, Mildorf)

Let $a, b, c > 0$. Prove that

$$(ab + ac + bc) \left(\frac{1}{(a+b)^2} + \frac{1}{(a+c)^2} + \frac{1}{(b+c)^2} \right) \geq \frac{9}{4}$$

We see that the inequality is symmetric, which is good, and that equality occurs when all variables are equal. So we can get started. I do not recommend to actually multiply out everything but to use σ_1 :

$$\begin{aligned} & (ab + ac + bc) \left(\frac{1}{(a+b)^2} + \frac{1}{(a+c)^2} + \frac{1}{(b+c)^2} \right) \geq \frac{9}{4} \\ \iff & 4\sigma_2 \sum_{\text{cyc}}^3 \left((a+b)(a+c) \right)^2 \geq 9 \left((\sigma_1 - a)(\sigma_1 - b)(\sigma_1 - c) \right)^2 \\ \iff & 4\sigma_2 \sum_{\text{cyc}}^3 (a^2 + \sigma_2)^2 \geq 9(\sigma_1^3 - \sigma_1^3 + \sigma_1\sigma_2 - \sigma_3)^2 \\ \iff & 4\sigma_2 \left(\sum_{\text{cyc}}^3 a^4 + 2\sigma_2(\sigma_1^2 - 2\sigma_2) + 3\sigma_2^2 \right) \geq 9(\sigma_1^2\sigma_2^2 - 2\sigma_1\sigma_2\sigma_3 + \sigma_3^2) \\ \iff & 4\sigma_2 \left(((\sigma_1^2 - 2\sigma_2)^2 - 2(\sigma_2^2 - 2\sigma_1\sigma_3)) + 2\sigma_2(\sigma_1^2 - 2\sigma_2) + 3\sigma_2^2 \right) \geq 9(\sigma_1^2\sigma_2^2 - 2\sigma_1\sigma_2\sigma_3 + \sigma_3^2) \\ \iff & 4\sigma_1^4\sigma_2 - 17\sigma_1^2\sigma_2^2 + 4\sigma_2^3 + 34\sigma_1\sigma_2\sigma_3 - 9\sigma_3^2 \geq 0 \\ \iff & 972d_1^4d_2 - 1377d_1^2d_2^2 + 108d_2^3 + 306d_1d_2d_3 - 9d_3^2 \geq 0 \end{aligned}$$

and here we run into some difficulties. Of course, the d_3^2 is not a problem, it is smaller than any other summand. We could do $108d_2^3 \geq 108d_1d_2d_3$ and then we are left to show

$$972d_1^4d_2 + 414d_1d_2d_3 \geq 1368d_1^2d_2^2.$$

It would be nice if we had $972 : 414 \geq 3$, because in this case we could apply Schur's inequality, but unfortunately, this does not work, so we will have to come up with something better.

Enter the uvw method. Let us inspect

$$F(d_1, d_2, d_3) = 972d_1^4d_2 - 1377d_1^2d_2^2 + 108d_2^3 + 306d_1d_2d_3 - 9d_3^2$$

and note that we have to prove that $F(u, v, w) \geq 0$ whenever the zeroes of the polynomial $P(t) = t^3 - 3ut^2 + 3vt - w$ are positive real. Call (u, v, w) *admissible* in that case.

Note that

$$\frac{\partial F}{\partial w} = 306uv - 18w > 0^{15}.$$

This means, if $w = d_3$ is getting smaller, the inequality only becomes sharper. So, if the inequality holds for $(d_1, d_2, d_3 - \varepsilon)$, then it holds for (d_1, d_2, d_3) as well. So, we can decrease d_3 to the lowest value so that (d_1, d_2, d_3) is still admissible. But when does it occur? Let us look at the graph of $P(t)$:

¹⁵The funny differential sign on the left side of the equation denotes a partial derivative. This just means derivative of F with respect to w . That is, treat u and v as parameters and w as your variable. If you are uncomfortable with derivatives, feel free to check that if we increase w , $F(u, v, w)$ increases as well (as long as (u, v, w) is admissible).

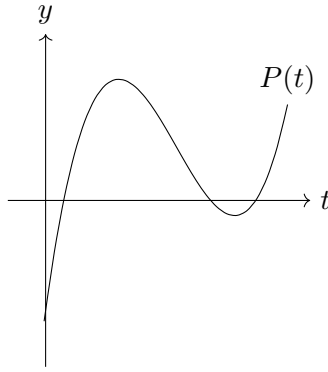


Figure 7: $P(t)$ has three positive zeroes, corresponding to a , b and c .

We see that decreasing d_3 just moves the graph upwards (as $P(t)$ has a $-d_3$ term). When do the zeroes stop being positive? Basically, two things can happen. Either, the intersection with the y -axis reaches the origin. In that case, one of the variables is zero. In the other case, the two larger zeroes coincide (shortly before they vanish).

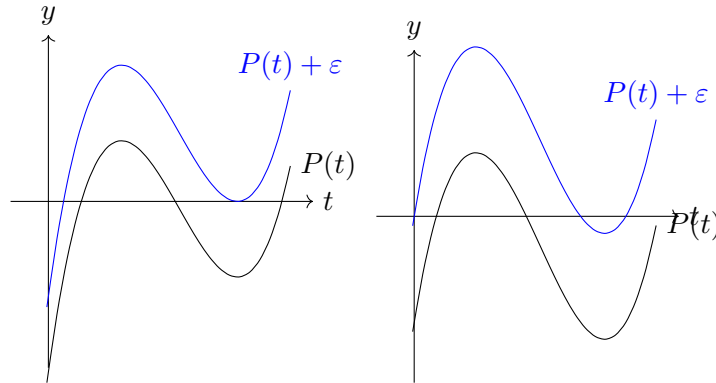


Figure 8: One of two things will happen if we increase f slightly (or equivalently decrease d_3 slightly). Either, as in the left, two variables will become the same. Alternatively, the smallest variable will be zero, as on the right, where one of the zeroes of $P(t)$ reaches the origin.

So, let d'_3 be the lowest value such that (d_1, d_2, d'_3) is still admissible. We only need to prove that $F(d_1, d_2, d'_3) \geq 0$. But (d_1, d_2, d'_3) corresponds to a set of variables (a', b', c') . We write a, b, c instead of a', b', c' for brevity. As the inequality is symmetric, WLOG assume $a \leq b \leq c$. So, because of the minimality of d'_3 , we can assume that either $a = 0$ or $b = c$. But those cases are not hard.

If $a = 0$, the remaining inequality is still homogenous. So, let us assume $b = 1$ for further simplification. So, we only have to prove that

$$\begin{aligned}
 & c \left(1 + \frac{1}{c^2} + \frac{1}{(1+c)^2} \right) \geq \frac{9}{4} \\
 \iff & 4(c^2(c+1)^2 + (c+1)^2 + c^2) \geq 9c(c+1)^2 \\
 \iff & 4c^4 - c^3 - 6c^2 - c + 4 \geq 0 \\
 \iff & (c-1)(4c^3 + 3c^2 - 3c - 4) \geq 0 \\
 \iff & (c-1)^2(4c^2 + 7c + 4) \geq 0,
 \end{aligned}$$

which is true.¹⁶ You can check the case $b = c$ analogously for yourself. The easiest way is to say that WLOG, $b = c = 1$.

Remark 7.18. If you are afraid of the fact that the original problem only asks for positive reals, note that the above argument generalises the problem for $a \geq 0$ and $b, c > 0$. If the problem has positive reals and $\geq$ instead of $>$, setting something to zero will never be a problem. (You might divide by zero, in this case the only thing that happens is that you generalise the multiplied-out inequality.)

Warning 7.19. Graders usually don't like it if you say "increase until" and they prefer "increase by". This is because it is unclear what the former really means on a rigorous level (i.e. does there exist such a smallest d'_3 etc.). Fortunately, it is not a big deal for you to say that you increase the constant term by $\min(-d_3, -\eta) \geq 0$, where $\eta \leq 0$ is the minimum of $P(t)$ in the interval $[b, c]$. You can state in contest that $P(t)$ attains a minimum on the compact interval $[b, c]$.

Remark 7.20. You can just as well increase/decrease d_1 or d_2 , but it is harder to state that formally. I recommend not using the polynomial $P(t) = t^3 - 3ut^2 + 3vt - w$, but $P_1(t) = t^2 - 3ut + 3v - \frac{w}{t}$ and $P_2(t) = t - 3u + \frac{3v}{t} - \frac{w}{t^2}$. Those functions have the same zeroes as $P(t)$ and their minima/maxima on suitable intervals tell you how much you should increase or decrease d_1 or d_2 , respectively.

With this, we can describe perfectly what our general strategy is:

1. Write the inequality in terms of the ESP. If it dies by Maclaurin, Newton and Schur, you are done. Otherwise proceed as follows:
2. Find one ESP in which the inequality is monotonous.
3. Set this ESP to the minimal/maximal value such that (d_1, d_2, d_3) is still admissible.
4. Exploit that WLOG, one of the variables is zero or two variables are equal.

Try the uvw method on the following problems:

Problem 7.21 (Problem 4.3, Cauchy-Schwarz'ed). Let $a, b, c > 0$ be real numbers. Prove that

$$(ab + ac + bc) \left(\frac{1}{(2a + b + c)^2} + \frac{1}{(a + 2b + c)^2} + \frac{1}{(a + b + 2c)^2} \right) \leq \frac{9}{16}$$

Hints: 89

Problem 7.22 (German MO 2015/6). Let a, b, c be positive real numbers. Prove that

$$\frac{a + b + c}{3} + \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \geq \sqrt[3]{\frac{abc}{16}}$$

Hints: 182

¹⁶One-variable inequalities with equality case at p almost always work by factoring $(x - p)^2$ out and then hoping that everything is positive.

Problem 7.23. Let a, b, c be nonnegative real numbers with $a^2 + b^2 + c^2 + abc = 4$. Prove that

$$a + b + c \geq 2 + \sqrt{abc}$$

Hints: [63](#) [98](#)

For more about the uvw method, see [\[4\]](#).

§8 More Problems

Problem 8.1 (Polish MO Second round 2024 P5). The positive reals a, b, c, x, y, z satisfy

$$5a + 4b + 3c = 5x + 4y + 3z.$$

Show that

$$\frac{a^5}{x^4} + \frac{b^4}{y^3} + \frac{c^3}{z^2} \geq x + y + z.$$

Hints: [135](#) [157](#)

Problem 8.2 (Baltic Way 2025/5). Let n be a positive integer and let $x_1 \leq x_2 \leq \dots \leq x_n$ be positive real numbers satisfying

$$\frac{x_1^3 + x_2^3 + \dots + x_n^3}{x_1 + x_2 + \dots + x_n} = \frac{x_1^2 + x_n^2}{2}.$$

Prove that

$$x_1^2 + x_2^2 + \dots + x_n^2 \geq nx_1x_n.$$

Hints: [136](#) [70](#) [107](#) [149](#)

Problem 8.3 (German MO 2021/5). Find the largest real number A such that

$$\frac{1 + yz}{1 + x^2} + \frac{1 + xz}{1 + y^2} + \frac{1 + xy}{1 + z^2} \geq A$$

for all real numbers x, y, z . Hints: [79](#) [74](#) [43](#) [148](#)

Problem 8.4 (ISL 2017/A1). Let $a_1, a_2, \dots, a_n, k$, and M be positive integers such that

$$\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} = k \quad \text{and} \quad a_1 a_2 \dots a_n = M.$$

If $M > 1$, prove that the polynomial

$$P(x) = M(x + 1)^k - (x + a_1)(x + a_2) \dots (x + a_n)$$

has no positive roots. Hints: [55](#) [53](#) [48](#)

Problem 8.5 (ISL 2019/A4). Let $n \geq 2$ be a positive integer and $a_1, a_2, \dots, a_n$ be real numbers such that

$$a_1 + a_2 + \dots + a_n = 0.$$

Define the set A by

$$A = \{(i, j) \mid 1 \leq i < j \leq n, |a_i - a_j| \geq 1\}$$

Prove that, if A is not empty, then

$$\sum_{(i,j) \in A} a_i a_j < 0.$$

Hints: [118](#) [57](#) [56](#) [20](#)

Problem 8.6 (IMO 2020/2). The real numbers a, b, c, d are such that $a \geq b \geq c \geq d > 0$ and $a + b + c + d = 1$. Prove that

$$(a + 2b + 3c + 4d)a^a b^b c^c d^d < 1.$$

Hints: [99](#) [146](#)

§9 Hints

1. Find correct weights.
2. It is okay not to use up all two units of $x^2 + y^2 + z^2$. After all, $x^2 + y^2 + z^2 \geq (xy + xz + yz) = 1$.
3. You might have to apply AM-GM multiple times.
4. What if one of the variables is zero?
5. Find weights.
6. Try to prove that g can only attain a minimum where $x_i + x_j = 0$ for some i, j .
7. Go through the summands on the RHS. What weights will you need?
8. $y + z = 1 - x$. How to produce $y + z$?
9. Denominator nullification.
10. What do you know about the polynomial with zeroes a, b, c, d ?
11. Define y_i such that the condition tells something about $y_i - y_{i-1}$.
12. Apply what you just found out to the denominators on the RHS **very carefully**
13. You can subtract each summand from something that is not a constant.
14. What is “the function” here?
15. Apply AM-GM on each factor with appropriate weights.
16. To respect the equality case $x_1 = x_2$, you will have to have $a + b = \frac{1}{3}$.
17. There exists an extremely short solution to this problem.
18. $f(x) = \ln(x^2 + 1)$.
19. Actually, the inequality has become an equality now.
20. After removing all $a_i > 1$ and $a_i < -1$, you can look at *all* products.
21. Find P, Q, R such that $PQR = \frac{a}{b+7}$ and the sums $\sum P$, $\sum Q$ and $\sum R$ are manageable.
22. Apply Cauchy-Schwarz in a clever way.
23. Finish off by a suitable use of AM-GM. Keep the equality case in mind.
24. One of the numerators $2^{k-1} + 1, 2^{k-1} + 2, \dots, 2^k$ will have to increase the sum.
25. Increase a_1 and decrease a_3 .
26. Alternatively, consider $(x - 1)^2(y - 1)^2 \geq 0$.
27. The equality case occurs for e.g. $a = 49b$. What can you choose as Q and R , given the equality case of Hölder?
28. You can not only have sums, you can also have products, as long as they are symmetric expressions in x, y, z .
29. Estimate each summand against something similar to the RHS.

30. Use an exponential function.
31. Try to find the equality case. This should tell you which tangent to use.
32. First, make a substitution, so you can apply $n - 1$ EV.
33. This should let you remove one variable. Induct.
34. $P = \frac{a}{a+7}$
35. To find minima/maxima of a one-variable function, differentiate it.
36. Does the telescope still work, when the denominator is there?
37. Multiply out the square and apply Cauchy-Schwarz again.
38. ...or alternatively, get completely rid of it by starting with subtracting 3 from the inequality at the very beginning.
39. All derivatives vanish. Try every derivative when you increase one variable and decrease an other.
40. This problem has very few degrees of freedom.
41. Wrap up from right to left. (I mean from a_n to a_1 , not necessarily to start with the RHS.)
42. $(\frac{2}{1+x})^2 + (\frac{2}{1+y})^2 \geq 2 \cdot \frac{2}{1+xy}$
43. Make a WLOG size assumption and exploit it carefully.
44. Apply Hölder.
45. The square root and $x \mapsto \sqrt{-x}$ are both concave. Draw some pictures.
46. Look at the polynomial with zeroes a, b, c, d . What do you want from it?
47. Which side is bigger?
48. Secretly, it is a one-variable inequality.
49. Apply Titu's lemma on the given condition to extract that $a + b + c \leq 3d$.
50. Multiply the inequality you want to show by $(a + 1)$.
51. $\int_0^1 x^2 dx = \frac{1}{3}$.
52. You can even make a WLOG assumption.
53. Insert the values of k and M into the inequality.
54. What is the "equality case"?
55. Is P positive or negative?
56. What can you say about variables a_i with $|a_i| \geq 1$?
57. Look at the complement of A instead.
58. The inequality is not really sharp: You can prove that $a^2 + bc \geq \frac{a^2+b^2+c^2}{2}$
59. Apply Cauchy-Schwarz to $(x^4 + y)(z + y^4)$.
60. Multiply out.
61. Keep $a_1 + \dots + a_n$ fixed. Move something.
62. Subtract the first summand from a .
63. The problem is that you have a condition which uses all three ESP. So you cannot just move one of them. Is there a way to fix this?
64. Treat the x^5 and x^2 parts differently. For one of them, you can apply denominator nullification...

65. Morally, the sum of squares is huge. How do you create a fair condition?
66. Use the Schur inequality carefully. What happened to the equality cases?
67. $\frac{a}{b} + \frac{c}{d} \geq 2\sqrt{\frac{ac}{bd}}$.
68. Each summand should become a big square root. Apply AM-GM again, very carefully.
69. Feel free to expand a fraction to get a square in the numerator.
70. What does it mean algebraically that $x_1 \leq x_i \leq x_n$? Note that in the inequality you want to show, x_1 and x_n play the same role.
71. Try to produce $ab + ac + bc$ somewhere.
72. If all x_i are as in the “equality case”, then neither the condition is satisfied, nor is the inequality sharp. It is quite close though.
73. Can you relate the sum to the integral, when you ignore the $1 - a_k$ in the denominator?
74. Try to send the variables to zero or infinity in a clever way.
75. Try to prove the telescope by pure algebraic manipulation.
76. Find squares in the numerators.
77. Prove the lower bound in case of $n = 2^k$ inductively.
78. You will have to find weird AM-GM weights.
79. The equality case is very weird.
80. Rewrite the condition in terms of ESP.
81. Write down the associated polynomial. The problem asks you to prove that all three zeroes of it are sufficiently large - go for it!
82. Equality occurs if $x + 2y + 3z = 0$.
83. You don't have to grab the entire summand.
84. $n - 1$ EV. You will only need to consider $a < 1 < b, c$ (why?)
85. How to write each summand as a function of only one variable? Remember, you will have to keep the *sum* of the variables fixed.
86. $\frac{x}{x+7} + 7\frac{1}{x+7} = 1$
87. Use $\frac{a^r}{a^r + b^r + c^r}$.
88. As $xyz \geq 1$, feel free to move monomials in and out.
89. I'm afraid this is going to be quite brutal...
90. Would you rather push x and y together or apart?
91. The factors on the LHS are huge. Estimate each of them against your favourite polynomial $f(a)$ with low degree and $f(1) = 3$.
92. $f(a)$ should have “average degree” 1. Still, try to keep $f(a)$ large.
93. Secretly, this is a one-variable inequality.
94. $a^2 + b^2 \geq 2ab$.
95. What has to be the case if x is a zero of this polynomial?
96. And at the very end, use calculus once more to get rid of the last variable.
97. The constraint tells you that $ac + bd$ is quite large. Try to create it somewhere.
98. Turn the table around.

99. The problem authors felt funny and put variables in the exponents. What do you do in such a case?
100. This problem is very evil because it is very sharp. Most estimates like $2\sqrt{ab} \leq a + b$ already give too much away.
101. Define $g(t)$ to be the RHS when the variables are $x_1 + t, x_2 + t, \dots, x_n + t$. Where can g attain a minimum?
102. Estimate against $ka^2 + (\frac{2}{3} - k)b^2$.
103. Look at the polynomial with zeroes x, y, z carefully.
104. Now, something funny happens because of $abc = 1$.
105. Estimate each summand against something like $ax_1 + bx_2$.
106. Try to write $\sqrt{x}$ as an AM-GM of $\frac{1}{x}$ -like and xy -like things.
107. $(x_i - x_1)(x_i - x_n) \leq 0$.
108. Remove the variables one-by-one. Be careful!
109. It is hard to guess the equality case. It has manageable numbers though...
110. $4(a^3 + b^3 + c^3)$ is huge and 3 of it is obsolete. What do you do with it?
111. Compare the inequality you want to prove with the condition given. You might want to do a proof by contradiction.
112. Find appropriate AM-GM weights.
113. If n is even, you can manage the sum $x_1x_2 + x_2x_3 + \dots + x_nx_1$ quite well.
114. Try to create $1 + x_i$ in each numerator.
115. The equality case is weird. Morally, a and c are large, while b and d are small.
116. Do something very subtle in the denominator.
117. Mimic the proof presented above.
118. How do you use the sum condition?
119. Use $\frac{1}{x} + y^2 + z^2$.
120. Do not let this problem fool you! x, y and z can be negative!
121. Relate x_i, x_i^2 and ab for each i .
122. "Solve" the condition first.
123. This is an AM-GM problem. First, focus on the part that has $(x - y)$ as a factor.
124. If one of the variables is zero, you have a lot of linear functions. You can say that WLOG, more variables are zero.
125. $x_1^l x_2^l \dots x_n^l = 1$.
126. Try to find a function f such that $f(a_1) + \dots + f(a_n)$ behaves nicely.
127. Use $\frac{a}{x} + \frac{b}{x} + \frac{c}{y}$.
128. Try to extract something more useful from the given condition.
129. What if $AB = CD, AC = BD$ and $AD = BC$?
130. If the denominator on the RHS was $(1 + a_1)(1 + a_2)(\dots)(1 + a_n)$ instead, what would you do, step by step, right to left?
131. In general, which side is bigger? Try to find equality cases.

132. The computation at the end is just tedious. I'm sorry.
133. Try to write the RHS as a telescope.
134. Differentiate the RHS with respect to n .
135. Find a telescope - split different variables.
136. Build a telescope. Use x_1 , x_i and x_n to make an inequality.
137. Prove that $\frac{1}{a} \geq \frac{c}{a} + 2b$.
138. Subtract from $\frac{1}{2}$ each.
139. Use AM-GM to estimate $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ against an appropriate power/factor of $((x - y)\sqrt{xy})$.
140. Prove that the answer is monotonous in n .
141. Bring everything to a common denominator. Find a perfect square.
142. By Cauchy-Schwarz, $\sum \frac{b^2}{2a+b^2} \sum \frac{2a+b^2}{a} \geq (\sum \frac{b}{\sqrt{a}})^2$. The motivation for this is that $2a + b^2$ is roughly on the way from b^2 to something like $4a$.
143. This is more like a Cauchy-Schwarz puzzle than an AM-GM puzzle.
144. Estimate xy together with something like $\frac{a}{x} + \frac{b}{y}$.
145. $y_i = (\alpha x_{i+1} + \beta x_i)^2$
146. Just multiply out and estimate term by term.
147. Let $f(x) = x^4 - x^5$. Treat the convex and concave part differently.
148. Drop one of the three summands.
149. You have two versions of the inequality from the last hint - the normal one and the one where you have multiplied with x_i . Sum both up cyclically.
150. Solve a huge, and I mean huge, system of equations.
151. Try to make use of compactness.
152. This is a troll problem.
153. Split the denominators into three different inequalities.
154. Split up each fraction into the x^3 part and the y^3 part. Then, proceed as usual.
155. What could be a good telescope? Keep the product condition in mind.
156. Carefully inspect the equality case.
157. Show that $\frac{a^5}{x^4} - x \geq 5(a - x)$.
158. Try to find an estimate for $\frac{4}{a} + \frac{4}{b} + \frac{1}{c} + \frac{3}{d}$ and sum cyclically.
159. Homogenise.
160. Using this, you can now remove one variable.
161. Do the Ravi substitution and inspect the statement.
162. Apply AM-GM in the denominator carefully.
163. You will have to apply Hölder's inequality.
164. Rewrite each fraction as a function in one variable.
165. Feel free to use a slightly weaker form of the condition. You don't have to pay much attention to being extremely sharp, so try to keep things simple.

166. For the x^5 part, approach it directly...
167. Some constants urgently need to be replaced by $xy + xz + yz$.
168. On the boundary, one of the variables is zero. So, you are only interested in the case when the maximum is attained in the interior. Then, all derivatives are zero.
169. Add suitable multiples of $(x_i - x_{i-1})^2$ and $(x_{i+1} - x_i)^2$
170. Homogenise.
171. Prove that $(x+1)^2(y+1)^2 \geq 4(x^3 + y^3)$, whenever $|x - y| \leq 1$
172. For example, you can multiply the first line by z and add up cyclically. Find two more such equations.
173. You will need Schur's inequality.
174. Try small cases. Where is the equality case? Spoiler: you can do way better than n .
175. Create *symmetric* equations by combining the given information in a clever way.
176. Build your lemma that eliminates the last variable, similar to AM-GM. However, you want $1 + a_1 \dots a_n$, not $1 + a_1 \dots a_{n-1}(1 + a_n)$.
177. Write each summand as a one-variable function.
178. Go through the terms on the right side.
179. The function is convex and monotonous.
180. Use something like $\frac{x_1}{x_2}$ as a telescope.
181. After seeing that two variables are equal, you will have to use calculus again.
182. You can use monotonicity in *any* of d_1 , d_2 and d_3 .

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